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CHROMATE CONVERSION

THE COMPLETE TRAINING MANUAL

CONVERSION COATING · CHROMATE ON ZINC · CR^{3+} / ROHS-COMPLIANT

*A full training course for the brand-new operator — from "what even is a passivate?" to a signed certificate of completion. The modern, **RoHS-compliant** clear/blue and black **trivalent passivates** that finish freshly zinc-plated parts: the conversion film built on **trivalent chromium (Cr III)** instead of the old hexavalent chemistry, adopted across automotive, electronics, and fasteners because it contains **no hexavalent chromium.** No electricity — just chemistry, plus the topcoat/seal that makes the system perform. Assumes you know nothing.*

01 RECEIVE ▸ 02 ACTIVATE ▸ 03 RINSE ▸ 04 PASSIVATE

05 RINSE ▸ 06 TOPCOAT·SEAL ▸ 07 DRY·CURE ▸ 08 INSPECT

OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 · 2026 · BEGINNER-FRIENDLY STYLE

Training reference only. **Trivalent chromium (Cr III) is the RoHS / ELV / REACH-compliant chromate chemistry** and is far lower-hazard than hexavalent chromium — but it is still an **acidic chemical process**, and many topcoats/sealers carry their own hazards (e.g., cobalt). Always verify exact parameters against your supplier's TDS, customer specs, and current SDS, and comply with OSHA, EPA, REACH/RoHS/ELV, and local regulations before production use. **Regulatory data reviewed 2026-07-06:** exposure limits reflect published U.S. federal OSHA PELs as of this date; referenced standards should be checked against their current revision, and state limits (e.g., Cal/OSHA) and ACGIH TLVs may be stricter.

PART ONE — FRONT MATTER & FUNDAMENTALS
START HERE

WELCOME ABOARD

IF YOU DON'T KNOW A PASSIVATE FROM A PAPERWEIGHT, YOU'RE IN EXACTLY THE RIGHT PLACE — AND HERE YOU'RE LEARNING THE *MODERN, COMPLIANT* WAY TO FINISH ZINC

Welcome. If you're holding this manual, you're about to learn how to run — or at least understand — a **trivalent chromate (passivate) line**: the quick chemical dip, plus a topcoat, that goes on **freshly zinc-plated parts** to turn bright, bare zinc into a corrosion-resistant, paint-ready finish. Maybe you started this week; maybe you've pulled baskets off the zinc line for a year and want to know *why* that last dip-and-topcoat matters. Either way, you're in the right place.

Here is the most important thing we can tell you up front, and we will say it more than once: **trivalent chromate is the RoHS-compliant modern default for passivating zinc — and it is built on trivalent chromium (Cr III), which is *not* a carcinogen.** It is far lower-hazard than the old hexavalent (Cr VI) chemistry it replaced. But "lower-hazard" is not "no-hazard": it's still a warm, **acidic chemical bath**, the topcoats can contain sensitizers like **cobalt**, and you still need ventilation, PPE, and good housekeeping. The second thing: **this manual assumes you know nothing about chemistry, zinc plating, or how a "passivate" works.** That's a promise, not an insult: every term gets defined the first time we use it, in the spirit of the *plain-language* guides — friendly and patient.

> ★ **REMEMBER**
> You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday — and on a finishing line, the understanding is what keeps both the parts and you in good shape.

> ⚠ **HEADS-UP**
> One big idea before we start: **a chromate line has no electricity running through the parts.** The coating forms because the zinc surface and the passivate solution *react with each other.* That removes the shock-at-the-part hazard. It does **not** make the line hazard-free: you are still working with **acidic activation and passivate dips, cobalt-bearing topcoats, warm drying/curing air, and slick wet floors.** Respect all of it.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, in the order a part travels — receive, activate, rinse, passivate, rinse, topcoat/seal, dry/cure, inspect. Reading it in order matters: the order of a finishing line is never random, and neither is this book.

> 💡 **TIP**
> Read Part One all the way through *before* any individual station chapter — the station chapters assume you already understand the big picture (especially the **passivate + topcoat system** and why trivalent isn't self-healing) that Part One gives you.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

> 💡 **TIP**
> A shortcut or trick of the trade — the kind of thing a 20-year veteran would lean over and tell you.

> ★ **REMEMBER**
> A core idea worth locking in. If you forget everything else on the page, don't forget these — they're the load-bearing ideas.

> ⚠ **HEADS-UP**
> A safety warning. Read every single one. On a trivalent line these keep your skin, eyes, lungs, and the part's quality intact. The hazards are lower than on a Cr VI line — but they're real.

> ⚙ **TECHNICAL STUFF**
> Deeper chemistry for the curious. Optional — skip it and you can still run the line.

> ☀ **FUN FACT**
> A bit of trivia to make the science stick (and give you something to say at lunch).

TWO THINGS YOU'LL SEE AT EVERY STATION

Each station chapter ends with a **Teach-Back** checklist (the things you must say out loud, unprompted, before you're cleared) and a bordered **Sign-Off** box your *trainer initials* once you've demonstrated the skill safely. Teach-Back is your study guide; Sign-Off is the gate.

FIND YOUR WAY

TABLE OF CONTENTS

READ FRONT-TO-BACK THE FIRST TIME · KEEP ON THE SHELF FOREVER AFTER

★ **REMEMBER**

This manual is a training and reference tool — **not** a substitute for your facility's written procedures, your supplier's TDS/SDS, or your supervisor's instructions. When this manual and your shop's official procedure disagree, **follow the official procedure and ask your supervisor.**

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS FOUNDATION AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what a conversion coating is** in plain language — and how it differs from electroplating and anodizing (chemistry alone, *no electric current*).
2. **Describe what a trivalent chromate (passivate) is and what it does** — a thin chromium/zinc conversion film grown on freshly zinc-plated parts that boosts corrosion resistance, adds color, and serves as a paint base — built on **trivalent chromium (Cr III), with no hexavalent chromium.**
3. **State the central technical fact of this chemistry:** trivalent passivate is a **passive barrier film — it is NOT self-healing** like old hex chromate. To match hex's corrosion performance, trivalent relies on a **thicker, more uniform passivate plus a topcoat/seal** — the "passivate + topcoat system."
4. **State that the passivate is the *finishing step after zinc plating*** — the "passivate" the zinc manuals hand off to — and name the substrates it works on (zinc, zinc-nickel, zinc-iron, zinc-cobalt, cadmium).
5. **Read the trivalent color palette** — clear/blue-bright (thin film), iridescent, and **black** (thicker film, usually with a topcoat) — and understand that color comes from **film + topcoat**, not from a leachable chromate.
6. **State the safety facts of this line:** Cr(III) is **not a carcinogen** and is far lower-hazard than Cr(VI), but the bath is still an **acidic irritant**, some **topcoats contain cobalt (a sensitizer)**, and you must **not let the trivalent chemistry oxidize to or be cross-contaminated with hexavalent.**
7. **Explain the regulatory driver** — that **RoHS, ELV, and REACH restrict hexavalent chromium**, which is exactly why trivalent (Cr III) became the **modern default** for zinc passivation.
8. **Name the stations of the line in order** and explain *why* the order can't be rearranged.
9. **Apply the key operator process lessons** — run the passivate to the right film, *don't skip or shortchange the topcoat,* dry/cure per the TDS, and keep drag-in and oxidizers out of the trivalent bath.
10. **Spot trouble early** — low salt spray, no/thin color, topcoat defects, white rust — and talk the language with your lead, your chemist, and your supplier.

★ **REMEMBER**

You are not expected to hit all of these on day one. Surface finishing is a craft. But the *safety* objectives (#6) and the *system* objective (#3 — passivate + topcoat) are the two ideas everything else hangs on.

• THE EIGHT-STATION PROCESS AT A GLANCE

The whole trivalent line in one strip. Don't memorize it yet — just feel the rhythm: **Receive → Activate → Rinse → PASSIVATE → Rinse → TOPCOAT/SEAL → Dry/Cure → Inspect.**

01 **RECEIVE** zinc in → 02 **ACTIVATE** acid dip → 03 **RINSE** overflow → 04 **PASSIVATE** Cr III, pH ~1.8-2.5 → 05 **RINSE** controlled → 06 **TOPCOAT** seal step → 07 **DRY/CURE** per TDS → 08 **INSPECT** color & rub

★ **REMEMBER**

Notice the pattern: **a rinse follows each working stage**, and unlike the old hex line, the **topcoat/seal (Station 06) is a starring step, not an afterthought.** On trivalent, the passivate and the topcoat are a *team* — the passivate converts the surface, the topcoat delivers the corrosion numbers.

⚠ **HEADS-UP**

This eight-station layout is the *teaching* version. Your shop's real line may add or combine steps (e.g., a separate desmut, a dragout-recovery rinse, two topcoats, thick-film vs thin-film passivate options). Always run *your* line's actual posted sequence and parameters.

CHAPTER 1

WHAT IS A CONVERSION COATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT "FINISHING" MEANS

THE ONE-SENTENCE VERSION

A conversion coating is a protective film you grow on a metal surface by making the metal itself react with a chemical solution — turning the top layer of the metal *into* the coating. That's the whole idea. Now let's unpack it, because the word "conversion" is doing a lot of work.

THREE COUSINS: PLATING, ANODIZING, AND CONVERSION COATING

If you've seen our other manuals, you've met two other finishing families. It's worth lining all three up side by side, because the differences are the key to understanding chromates.

- **Electroplating** *adds* a layer of a *different* metal on top of your part, using **electricity.** (Zinc plating — the step right *before* this one — is electroplating: electricity drives dissolved zinc out of a bath and onto the steel.)
- **Anodizing** *converts* the part's *own* surface into a thick, hard oxide — but it still uses **electricity** to drive the reaction, and only on metals like aluminum.
- **Conversion coating** (this manual) *converts* the part's own surface into a thin protective film using **only chemistry — no electricity at all.** You dip the part, the surface reacts with the solution, and a new compound forms right where the bare metal used to be.

★ **REMEMBER**

Lock in the difference: **Plating adds a new metal with electricity. Anodizing converts the surface with electricity. A conversion coating converts the surface with *chemistry alone* — no rectifier, no anode, no cathode, no current.** If you go looking for the "power supply" feeding the parts on a passivate line, you won't find one. That's not a missing piece — that's the whole point.

WHERE THIS ONE FITS: AFTER PLATING, NOT INSTEAD OF IT

Here's the twist that makes chromate special among conversion coatings: **the surface it converts is itself an electroplated layer.** A chromate doesn't react with steel — it reacts with the **zinc** (or zinc-alloy, or cadmium) that was just plated *onto* the steel. So a trivalent passivate line is almost always the **tail end of a zinc-plating line:** plate the zinc, rinse it, then passivate it (and topcoat it). The zinc protects the steel; the passivate (plus topcoat) protects the zinc.

☀ **FUN FACT**

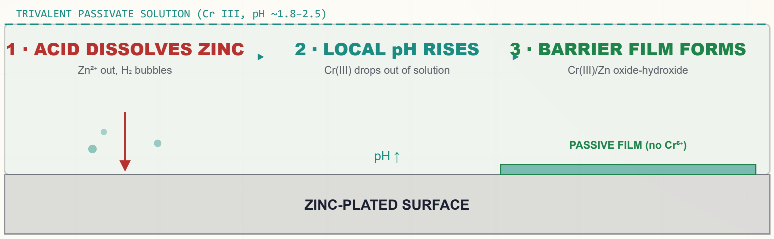
Chromate conversion coatings go back to the 1920s–1930s and ran on hexavalent chromium for most of a century. **Trivalent passivates are the modern chapter of that same story** — developed and scaled up especially from the 1990s–2000s onward, when RoHS and the automotive ELV directive made hexavalent chromium unacceptable for most commercial products. Same job, same family, safer chemistry.

HOW A CONVERSION COATING FORMS — NO CURRENT REQUIRED

So if there's no electricity pushing metal around, what makes the coating appear? The metal and the solution do it themselves. Here's the plain-language version for a trivalent passivate on zinc:

1. The passivate solution is **acidic** (typically mildly acidic, pH around 1.8–2.5, product-specific). When a freshly zinc-plated part touches it, the acid begins to **dissolve a little of the zinc surface** — zinc goes into solution and tiny hydrogen bubbles form. (You'll often see fine bubbling on parts in the dip — that's the reaction working.)
2. That dissolving reaction **uses up acid right at the surface, so the local pH climbs.**
3. When the local acidity drops past a tipping point, the **trivalent chromium (and the other metals in the bath, like cobalt) can no longer stay dissolved at the surface** — they **precipitate** as a thin, hydrated film of **chromium(III)/zinc oxides and hydroxides** right onto the part.
4. Crucially — and this is the big difference from hex — **there is no hexavalent chromium in the bath, so none gets trapped in the film.** The film is a *passive barrier,* not a reservoir of mobile, self-healing chromate. (Hold that thought — it's the heart of Chapter 2.)

HOW THE FILM FORMS (NO ELECTRICAL CELL, NO HEXAVALENT CHROMIUM)



No electricity — and no hexavalent chromium. The zinc and the solution do it themselves.

The result is an **integral** coating — chemically grown into the zinc surface, not glued on top — that is more corrosion-resistant than bare zinc and gives paint and topcoats something to grip.

TECHNICAL STUFF

Roughly: acid attacks zinc ($Zn + 2H^+ \rightarrow Zn^{2+} + H_2\uparrow$), consuming acid and raising the interfacial pH. The rising pH precipitates a gel of hydrated **Cr(III) oxide/hydroxide**, co-depositing zinc and other bath metals (commonly **cobalt** and others, per the product) that improve corrosion resistance. The whole film is trivalent from start to finish — **no Cr(VI) is present, formed, or stored.** That single fact is what makes the coating RoHS-compliant *and* what removes the self-healing trick that hex relied on.

HEADS-UP

"No electricity at the part" removes the shock-at-the-tank hazard — but the line's heaters, pumps, hoists, dryer, and ventilation are still serious electrical equipment, and the baths are still acidic. **Any maintenance still requires Lockout/Tagout (LOTO).** No current through the parts does not mean no hazard in the building.

CHAPTER 2

WHAT IS A TRIVALENT CHROMATE (PASSIVATE)?

THE THIN CONVERSION FILM THAT FINISHES A ZINC-PLATED PART — CLEAR/BLUE, IRIDESCENT, OR BLACK — AND WHY IT NEEDS A TOPCOAT

A PASSIVATE'S BIG JOBS: PROTECT, COLOR, AND BOND — AS PART OF A SYSTEM

Here's the headline, and it's one of the most important ideas in this manual: **a trivalent chromate (also called a "passivate" or "conversion coating") is the finishing step that takes a freshly zinc-plated part and makes it more corrosion-resistant — but unlike the old hexavalent chromate, it does this as a *passive barrier* and almost always works *together with a topcoat/seal.*** It does three jobs:

1. **Corrosion protection.** Bare electroplated zinc is shiny but corrodes quickly, forming powdery white "white rust." A trivalent passivate film slows that — and **with a topcoat,** it can match or exceed the old hex chromate's salt-spray life.
2. **Color.** Clear/blue-bright (thin film), iridescent, or black (thicker film, usually with a topcoat) — both for appearance and as a rough visual signal of film type.
3. **Paint/adhesive base.** The chemically stable film gives paint, powder, adhesives, and topcoats a better grip than bright bare zinc.

> ★ **REMEMBER**
> A passivate is **the finish the zinc line hands off to.** Zinc plating protects the steel; the passivate (plus topcoat) protects the *zinc* and gives the part its final color and corrosion rating. "Zinc plate" and "passivate + topcoat" are the halves of one modern finish — almost nobody ships bright, un-passivated zinc.

THE KEY DIFFERENCE FROM HEX: TRIVALENT IS NOT SELF-HEALING

This is *the* idea that separates trivalent from the old hex chromate, so we'll state it plainly. The famous trick of hexavalent chromate was **self-healing:** the hex film trapped a reservoir of soluble, mobile **Cr^{6+}** that could leach out and chemically re-protect a scratch. That mobile Cr^{6+} was both the magic *and* the carcinogen.

Trivalent passivate contains no hexavalent chromium — so it has no mobile chromate to migrate and patch damage. It is a *passive barrier* film, not an *active,* self-healing one. Scratch it down to bare zinc, and the trivalent film cannot re-grow over the scratch the way hex could.

> ★ **REMEMBER**
> **Hex = active (self-healing) protection. Trivalent = passive (barrier) protection.** This is the single biggest functional trade you make for RoHS compliance. Everything about how you run a trivalent line — thicker/more uniform passivate, the mandatory topcoat, careful process control — exists to make a *passive* film perform as well as the old *active* one.

SO HOW DOES TRIVALENT MATCH HEX? THE PASSIVATE + TOPCOAT SYSTEM

If a single trivalent dip can't self-heal, how do modern zinc fasteners and auto parts hit **120, 240, 500+ hours** of salt spray? Through a **system,** not a single film:

- **(a) A thicker, more uniform passivate.** Trivalent chemistries (especially "thick-film" iridescent and black types) build a more substantial, even barrier than a bare clear hex dip.
- **(b) A topcoat / sealer — very commonly required.** This is the part that does the heavy lifting on corrosion numbers — sealing the film's pores and microcracks, adding a barrier, and often carrying **corrosion inhibitors** (silica-based, organic/polymer, or cobalt-containing systems).
- **(c) Careful process control.** Trivalent baths are more sensitive than hex to pH, contamination, drag-in, and temperature.

> ★ **REMEMBER**
> **On trivalent, the topcoat is not optional polish — it is usually how you actually meet the spec.** On the old hex line the seal was a bonus on top of an already-strong self-healing film. On trivalent, the passivate + topcoat are a *team.* Skip or shortchange the topcoat and you will fail salt spray.

THE TRIVALENT COLOR PALETTE — AND WHERE COLOR COMES FROM

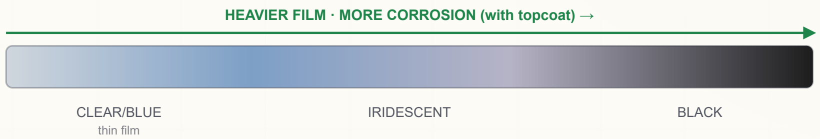
A trivalent passivate's color comes from a mix of **film type/thickness** and the **topcoat,** *not* from a trapped, leachable chromate. As a rough guide:

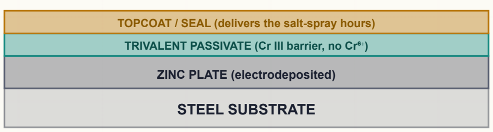
COLOR / TYPE	LOOK	TYPICAL FILM	NOTES
Clear / blue-bright	Nearly colorless, faint blue iridescence	Thin-film trivalent	The everyday decorative trivalent finish; usually topcoated for corrosion
Iridescent	Faint rainbow / pale shimmer	Thicker-film trivalent	Higher film weight, more corrosion; *not* the deep hex yellow-gold
Black	Deep black	Thicker-film trivalent + black topcoat	Decorative/functional black; color from the film + topcoat system

★ REMEMBER

Two ideas to lock in. **(1) Matching the old deep hex yellow/gold isn't the goal** — trivalent finishes are usually **clear/blue or black,** with iridescent in between. **(2) Color comes from film + topcoat, not from a leachable chromate** — so a trivalent's color is *not* a direct "fuel gauge" of protection the way the hex yellow was. Confirm the **exact color/type your job spec calls for,** and judge protection by *salt spray.*

READING A TRIVALENT FINISH · AND WHERE THE LAYERS SIT





The passivate is a passive barrier; the topcoat seals it and carries the corrosion performance.

⚙ TECHNICAL STUFF

In the old hex chromate, the yellow color literally *was* partly the trapped Cr(VI), so color and protection tracked together. Trivalent breaks that link: a clear/blue trivalent can out-perform an old clear hex once it's topcoated, and "more color" doesn't automatically mean "more protection," because there's nothing leachable to begin with. On trivalent you **specify and verify by salt spray and film type,** not by eyeballing the depth of yellow.

• WHERE TRIVALENT PASSIVATES WORK — THE SUBSTRATES

A passivate needs a reactive, freshly plated metal surface to convert. Trivalent passivates work on:

- **Electroplated zinc** — the classic and most common substrate (this manual's focus).
- **Zinc-alloy plating** — **zinc-nickel, zinc-iron, zinc-cobalt** — common for higher corrosion performance; these take trivalent passivates too (often with **alloy-specific** chemistries — zinc-nickel in particular usually needs a passivate formulated for it).
- **Cadmium** — historically chromated the same way (cadmium itself is now restricted in many applications).

> ⚠ **HEADS-UP**
> **A different metal needs a different conversion coating.** The chromate that finishes **aluminum** (the "chem film" / "Alodine" family, including trivalent aluminum chem films under **MIL-DTL-5541**) is a *separate process* with its own chemistry and is the subject of a different manual. This manual is **trivalent chromate on zinc and zinc alloys.** If aluminum parts show up on your line, flag it.

• WHERE TRIVALENT FITS IN THE SPECS

SPEC	WHAT IT COVERS	NOTE
ASTM B633	Electrodeposited **zinc on steel,** including supplementary finish **Types** and thickness **Classes**	The everyday workhorse spec; **current revisions include trivalent passivate options**
ASTM F1941 / F1941M	Electrodeposited coatings (zinc, etc.) on **threaded fasteners**	When the parts are screws, bolts, nuts; topcoat/torque-tension often specified here
Automotive OEM trivalent specs	Trivalent passivate + topcoat systems with required salt-spray hours	Many OEMs publish their own trivalent finish standards — run to the customer's exact callout
MIL-DTL-5541 (and 81706)	Chromate/chem-film on **aluminum** (incl. trivalent options)	**Different metal — separate manual.** Don't confuse with zinc passivate

> ⚠ **HEADS-UP — VERIFY THE EXACT TYPE/CLASS DESIGNATIONS**
> ASTM revises B633 periodically, and current editions have **added designations to cover trivalent passivates and to re-organize colored vs. colorless finishes.** Do **not** quote a Type number from memory on a job — pull the **current B633 revision** (and the customer's purchase-order callout, including any automotive OEM spec and required salt-spray hours) and confirm. We're flagging this rather than guessing a number that may be out of date.

> ★ **REMEMBER**
> **Class = how thick the zinc is. Type = what passivate/finish goes on top.** A modern spec also typically calls out **trivalent (Cr III / hex-free)** and a **required salt-spray performance** — and often a **topcoat.** Read the whole callout: zinc thickness, passivate type, topcoat, and the corrosion hours you must hit.

• THE HONEST TRADEOFFS

A good operator knows the weaknesses too:

- **It's not self-healing.** A scratch through to bare zinc won't re-protect itself the way hex would — which is exactly why the topcoat and good coverage matter.
- **It usually needs a topcoat to hit spec.** More steps, more chemistry, more cost than a bare hex dip — the price of compliance and safety.
- **The baths are more sensitive.** Trivalent passivates are fussier than hex about pH, temperature, drag-in, and contamination (including accidental oxidation toward Cr VI).
- **Color is different, not "matchable."** If a customer wants the exact old hex yellow look, set expectations — trivalent is a clear/blue or black world.

☀ **FUN FACT**

The faint blue or pale iridescence of a trivalent finish is mostly **thin-film interference** (the soap-bubble physics that colors an oil slick) over a near-colorless film — *not* the inherent color of a trapped chromate. You're reading film thickness with your eyes, not chromium content.

• A FIRST LOOK AT THE PROCESS NUMBERS

We'll go deep at Station 04, but here's the shape of the trivalent passivate dip so the rest of the manual makes sense:

PARAMETER	TYPICAL RANGE	WHY IT'S LIKE THIS
Bath type	Dilute acidic **trivalent chromium (Cr III) salts** + other metal salts (e.g., cobalt) + accelerators	The Cr(III) + acid + co-metals build the passive barrier film
pH	`~1.8–2.5 — product-specific; thick-film/iridescent types often higher (≈3–4)`	Acid drives the zinc-dissolving reaction that starts film growth
Temperature	`Room to warm (≈20–60°C / 70–140°F) — product-specific`	Trivalent often benefits from a little warmth to build film
Immersion time	`Short — ~30–60 s`	The reaction is fast; over- or under-immersion both hurt the film
Topcoat/seal	**Usually applied** as a separate dip after passivate + rinse	Where most of the corrosion performance comes from
Drying / cure	Warm dry/cure **per TDS** (topcoats often *require* a warm cure)	Cures the film/topcoat — but stay within the TDS limit

★ **REMEMBER**

Two things make a trivalent dip different from the old hex tank: it's often **a bit warmer and a touch less acidic,** and — most importantly — it's **half of a system.** The dip alone is not the finish; **the dip + topcoat + cure** is the finish. Don't think "chromate tank," think "passivate-and-topcoat system."

⚙ **TECHNICAL STUFF — CO-METALS AND ACCELERATORS**

Besides the trivalent chromium and the acid, trivalent baths contain **other metal salts (commonly cobalt, and others), fluorides, and accelerators/oxidizers** that control film build and corrosion resistance. **Cobalt** in particular is widely used to boost trivalent corrosion performance — and it's also why some topcoats/sealers carry cobalt-related hazards (Chapter 6). Exact identities and amounts come from the **supplier TDS**, maintained by your chemist/lab, not guessed at the tank.

CHAPTER 3 · THE MOST IMPORTANT CHAPTER

WHY TRIVALENT?

ROHS COMPLIANCE AND THE SWITCH FROM CR(VI), UP FRONT — READ THIS CHAPTER TWICE

We put this chapter near the front on purpose. The defining feature of a trivalent chromate is that it runs on **trivalent chromium, Cr(III)** instead of hexavalent — and that single change is why the entire industry adopted it. You need to understand both halves of the story: the *regulatory* driver (why the world demanded a hex-free passivate) and the *honest* hazard picture (much safer than hex — but not hazard-free).

WHY "TRIVALENT" IS THE WHOLE POINT

The old classic chromate ran on **hexavalent chromium (Cr VI)** — chromium in the **+6 oxidation state** — which is a **confirmed human carcinogen** and a strong, mobile oxidizer. Trivalent passivate is built instead on **chromium in the +3 state (Cr III).** That is the *same* element, in a *different,* far-less-hazardous chemical form — Cr(III) is even a trace dietary nutrient.

> ⚙ **TECHNICAL STUFF**
>
> "Hexavalent" (Cr^{6+}, Cr VI) and "trivalent" (Cr^{3+}, Cr III) describe the chromium atom's oxidation state. **Cr(VI) is a strong oxidizer, highly toxic, and carcinogenic. Cr(III)** is far less hazardous, poorly absorbed, and far less mobile. The whole design goal of a trivalent passivate is to **build the entire coating out of Cr(III) from the start — and keep it that way,** so neither the bath nor the finished part contains hexavalent chromium.

THE LAW: THE ROHS / ELV / REACH DRIVER

Three bodies of product-and-environmental law restrict hexavalent chromium in finished goods, and together they reshaped the whole industry:

- **RoHS** (Restriction of Hazardous Substances) — restricts Cr(VI) in **electrical and electronic equipment.**
- **ELV** (End-of-Life Vehicles directive) — restricts Cr(VI) in **automotive** parts.
- **REACH** — places hexavalent-chromium substances on the EU **authorization** list, meaning their use requires special, time-limited authorization and is being actively driven down.

The combined effect: **for most commercial markets — electronics, automotive, consumer goods, fasteners — hexavalent chromate is no longer acceptable.** Trivalent (Cr III) passivate is the chemistry the industry developed to replace it, and it is now **the default** for new commercial work.

> ★ **REMEMBER**
>
> The headline of this whole manual: **trivalent chromate exists because RoHS, ELV, and REACH restrict hexavalent chromium.** Trivalent is the **RoHS-compliant modern standard** for zinc passivation — chosen specifically because it contains **no hexavalent chromium.** That compliance story is *why your line runs trivalent.*

THE HONEST HAZARD PICTURE: SAFER, BUT NOT HAZARD-FREE

This is where we stay honest. Compared to the old hex line, a trivalent line is **far lower-hazard** — there's **no carcinogen in the main tank,** no need for the heavy Cr(VI) controls (no 5 µg/m³ air limit to chase, no Cr^{6+}-reduction waste step). But "safer" is not "safe," and a trained operator respects the real hazards that remain:

- **It's still an acidic chemical bath.** The activation and passivate baths are acidic and can cause **skin and eye irritation/burns.** Acid handling rules still apply.
- **Topcoats/sealers can contain cobalt — a sensitizer.** Many trivalent passivates and topcoats are **cobalt-bearing.** Cobalt is a recognized **skin and respiratory sensitizer,** and some cobalt compounds carry **carcinogen classifications** — *check the SDS for each product you run.*
- **Ventilation and PPE still apply.** You're working over warm, acidic, sometimes cobalt-containing tanks. Local ventilation and proper PPE are still required — just not at the extreme Cr(VI)-regulated level.
- **Don't let it oxidize to — or get cross-contaminated with — hexavalent.** Cr(III) *can* be oxidized to Cr(VI) under strongly oxidizing, hot, or high-pH conditions, or by drag-in of oxidizers. Letting that happen would re-introduce the carcinogen *and* break your RoHS compliance.

> ⚠ **HEADS-UP — THE ONE NEW "COMPLIANCE + SAFETY" RULE OF THIS LINE**
>
> **Keep the trivalent chemistry trivalent.** Don't drag oxidizers into the passivate bath, don't cross-contaminate with hex chromate chemistry or shared equipment, and follow the TDS on temperature and pH. If trivalent oxidizes to hexavalent, you've created a carcinogen *and* shipped a non-compliant part. Good housekeeping here protects people **and** the RoHS status of the finish.

> ⚙ **TECHNICAL STUFF — HOW CR(III) CAN BECOME CR(VI)**
>
> Oxidation of Cr(III) to Cr(VI) is favored by **strong oxidizers, high temperature, and high pH.** Practical drag-in culprits include strong **oxidizing acids and bleaches/peroxides.** This is one reason many trivalent lines use a **milder activation** (dilute sulfuric or a proprietary activator) rather than the strongly oxidizing **nitric** dip common on hex lines — to keep oxidizer drag-in out of the trivalent bath. Your TDS specifies the right activation; follow it.

> ⚠ **HEADS-UP**
>
> Don't let "trivalent is the safe one" make you sloppy. It's lower-hazard, not no-hazard: **acid burns, cobalt sensitization, and accidental oxidation to Cr(VI) are all real.** Respect the line every shift.

• HONEST POSITIONING: TRIVALENT VS. HEX

	TRIVALENT PASSIVATE (THIS MANUAL)	HEXAVALENT CHROMATE (PREVIOUS MANUAL)
Chromium state	**Cr(III) — far safer, not a carcinogen**	**Cr(VI) — carcinogen**
Self-healing	**No** — passive barrier; **needs topcoat/seal**	Yes — strong (mobile Cr^{6+})
Corrosion performance	**Excellent as a system** (passivate + topcoat)	Excellent on its own, cheaply
Cost / steps	More (topcoat usually required)	Less (bare dip can suffice)
RoHS / ELV / REACH	**Compliant — the modern standard**	**Restricted**
Status	**The default** for new commercial/automotive/electronics	Phasing out; survives in some defense/industrial

★

REMEMBER

Frame it honestly. **Trivalent is the RoHS-compliant, far-safer modern default — a passive-barrier film that usually needs a topcoat/seal to match the corrosion performance the old hex film gave on its own.** Hex was cheaper and self-healing and gave the best classic corrosion-per-dollar — but it's a carcinogen and RoHS-restricted, so it's being phased out. Both are legitimate to *know;* trivalent is the future.

☀

FUN FACT

Engineers spent years trying to give trivalent passivates "active" self-healing like hex. The practical answer the industry landed on wasn't a single miracle film — it was the **passivate + topcoat system,** sometimes with multiple sealer layers. It's a little like the difference between a coat that mends its own rips (hex) and a coat that's simply made of more, better-sealed layers (trivalent). Different strategy, comparable warmth.

⚠

HEADS-UP

If you take only one thing from this chapter: trivalent exists because hex is a **carcinogen** the world is **regulating out** of products. Trivalent gives you that compliance and a far safer line — but it only *stays* safe and compliant if you respect the acids, mind the cobalt, and **keep the chromium trivalent.**

CHAPTER 4

HOW A TRIVALENT LINE WORKS

A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS — AND ON TRIVALENT, THE PASSIVATE AND THE TOPCOAT ARE THE TEAM

A trivalent line is a **sequence of stations** a part travels through in a strict order. Because passivation is the *finishing* step, the line usually starts where the **zinc-plating line ends** — the part arrives already plated and rinsed. Parts are typically **barrel-processed** (small hardware/fasteners tumbled in a perforated barrel) or **rack-processed** (larger or cosmetic parts hung individually), the same way they were plated.

💡 **TIP**
The big mental model for this line: **everything before the passivate exists to deliver a chemically fresh, clean, active zinc surface; the passivate converts that surface; and the topcoat + cure deliver the corrosion numbers.** Unlike the old hex line, you can't think of the topcoat as optional — on trivalent it's where the performance lives.

• THE WHOLE TRIVALENT SEQUENCE (TEACHING LAYOUT)

#	STATION	THE ONE THING IT DOES	TEMP	NOTES
01	Receive & Inspect	Confirm the part arrived freshly **zinc-plated & rinsed**, in good condition	Ambient	Time-sensitive — fresh zinc passivates best
02	Acid Activation / Bright Dip	Quick dilute-acid dip to **desmut/brighten** the zinc and wake up the surface	Ambient	Often milder/non-oxidizing acid to protect the trivalent bath
03	Rinse (post-activation)	Wash off the activation acid	Ambient	The firewall before the passivate
04	**Trivalent Passivate**	**Grow the Cr(III) conversion film — the passivate**	Amb-warm	**Main conversion step; Cr(III); pH ~1.8–2.5; ~30–60 s**
05	Rinse (controlled)	Rinse off passivate drag-out without damaging the soft film	Cool/amb	Gentle; sets up the topcoat
06	**Topcoat / Seal**	**Add the topcoat/seal that delivers the corrosion performance**	Per product	**Usually required — silica, organic/polymer, often cobalt/inhibitors**
07	Dry-Off & Cure	Drive off water and **cure** the film/topcoat per TDS	Warm per TDS	Topcoats often *need* a warm cure — but stay under the TDS max
08	Inspect & Hand-Off	Confirm color/coverage and that the system is intact	Ambient	Fresh film/topcoat keeps hardening as it cures

The basic rhythm: **Receive → Activate → Rinse → PASSIVATE → Rinse → TOPCOAT/SEAL → Dry/Cure → Inspect.**

⚠ **HEADS-UP**
Two contrasts with the old hex line. **(1) The topcoat (Station 06) is a starring, usually-required step,** not an optional polish. **(2) Drying/curing is often *warm,* not cool-only** — many trivalent topcoats *require* a warm cure to perform — but there's still a TDS maximum above which you craze the film. Run *this* chemistry by *its* TDS; don't carry the hex "always dry cool" habit straight over.

• A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

★ **REMEMBER**

Every station either protects the next or poisons it. There is no neutral step. A part that arrives with stale, oxidized zinc won't passivate evenly. A part that drags activation acid (or an oxidizer) into the passivate shifts or contaminates the bath. A passivate run without a proper topcoat will fail salt spray. A topcoat dried wrong won't cure. What you do — or fail to do — early shows up as a corrosion failure later.

💡 **TIP**

Think of the line as building a *layered* defense: clean active zinc → trivalent conversion film → topcoat/seal → cure. The old hex line leaned on one clever self-healing film. The trivalent line wins by **stacking good, well-controlled layers.** Respect every layer.

• WHY "FRESH" ZINC MATTERS

Unlike steel, an **electroplated zinc surface ages.** Left sitting, bright zinc oxidizes and dulls within hours, and a dull, oxidized zinc surface **passivates poorly** — thin color, patchy coverage, weak protection. That's why a passivate line ideally runs **right after** the zinc line, and why the activation dip (Station 02) exists: to strip light oxide/smut and re-expose fresh, reactive zinc just before the passivate.

★ **REMEMBER**

Fresh zinc passivates well; stale zinc passivates badly. If parts have been sitting (overnight, over a weekend, or came in from another shop), expect to lean harder on activation — and flag heavily oxidized or white-rusted parts to your lead, because no passivate fixes badly corroded zinc.

☀ **FUN FACT**

The activation dip is sometimes called a "bright dip" because on a dull, slightly oxidized zinc part you can literally *watch* it flash back to clean, bright silver in the few seconds it's in the acid. That brightening is the smut and oxide dissolving away, re-exposing the reactive metal the passivate needs.

CHAPTER 5

WHY THE ORDER MATTERS

EACH STATION SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY — AND BY KEEPING THE CHROMIUM TRIVALENT

You might think the station order is just convention. It isn't. Each station exists *because of* the one before and after it.

RECEIVE FIRST — BECAUSE TIME AND CONDITION MATTER

Fresh, clean zinc passivates well; stale, oxidized, or contaminated zinc passivates badly. You confirm the part is genuinely freshly plated and rinsed, in good condition, *before* any passivate chemistry is wasted on a surface that can't accept it.

ACTIVATE BEFORE PASSIVATE — BECAUSE ZINC OXIDIZES

Even "fresh" zinc carries a light oxide/smut film that blocks an even reaction. The dilute-acid **activation/bright dip** dissolves that film and a whisker of zinc, leaving a chemically fresh, uniformly reactive surface. Skip it, and you get **patchy color and thin protection.**

> ★ **REMEMBER**
>
> **No activation, no uniform passivate.** A clean, freshly activated zinc surface is what lets the passivate form an even, full-coverage film. Activate right before you passivate — an activated surface re-oxidizes in minutes.

RINSE BETWEEN ACTIVATION AND PASSIVATE — BECAUSE DRAG-IN SHIFTS (AND CAN CONTAMINATE) THE BATH

The activation acid is its own chemistry; carry a film of it into the passivate and you **drift the passivate's pH and contaminate it** — and if the activation acid is a strong oxidizer (like nitric), drag-in can even nudge the bath toward forming **hexavalent** chromium. The rinse is the firewall.

> ★ **REMEMBER**
>
> *"Drag activation acid into the passivate and you blunt it — drag an oxidizer in and you risk making hex."* Rinses keep each chemistry out of the next tank. This is your first firewall on the line — and on trivalent it has a compliance job too: keeping the bath trivalent.

PASSIVATE — THE CONVERSION STEP (COOL-TO-WARM, QUICK)

Now — and only now, with a clean, freshly activated zinc surface — the trivalent film forms. The dip is **acidic, short (seconds), and run cool-to-warm per the TDS.** This is the conversion stage; treat it with respect, but remember it's only *half* the performance story.

> ⚠ **HEADS-UP**
>
> The passivate solution is **acidic** and may be **cobalt-bearing.** Even though it's only seconds of immersion, the mist, the drag-out, and any splash are acidic (and possibly cobalt-containing). Wear your PPE and keep ventilation running. (Full detail at Station 04.)

RINSE AFTER PASSIVATE — GENTLY

The part leaves the passivate wet with acidic drag-out that must come off before the topcoat — but the **fresh film is soft.** Rinse **gently and controlled** — enough to wash off drag-out and prep the surface for the topcoat, without scrubbing the soft film off.

> ★ **REMEMBER**
>
> This rinse's job is to **clean off drag-out and hand a clean film to the topcoat.** Skimp and you contaminate the topcoat tank; over-do it roughly and you damage the soft film. Clean, controlled, gentle.

TOPCOAT/SEAL, THEN DRY/CURE, THEN INSPECT

- **Topcoat/seal** — usually **required** on trivalent — adds the corrosion life, lubricity/torque-tension control, and (often) color. This is where you hit the salt-spray spec.
- **Dry and cure per TDS** — frequently a *warm* cure (unlike hex's cool-only rule), but never above the TDS max, which can craze the film.
- **Inspect last** to confirm color, coverage, and an intact, cured system.

> ⚠ **HEADS-UP — THE ORDER *AND* THE CURE ARE LOCKED**
>
> On trivalent, **the topcoat comes after a clean rinse and before the cure, and the cure follows the TDS.** Skip the topcoat and you fail salt spray; over-bake the cure and you craze the film. The sequence and the cure window are both part of the procedure, not suggestions.

> ⚙ **TECHNICAL STUFF**
>
> The recurring villain that justifies the rinses is **drag-out** (solution clinging to a part as it leaves a tank) and its mirror, **drag-in** (that film contaminating the next tank). On a trivalent line, drag-out is less acutely hazardous than on a Cr(VI) line, but it still wastes chemistry, spreads acidic/cobalt-bearing solution, contaminates downstream tanks, and loads your wastewater with metals. **Dragout-recovery rinses** and tidy handling keep the system clean and the topcoat tank uncontaminated.

CHAPTER 6

PPE — YOUR DAILY ARMOR

THE GEAR THAT STANDS BETWEEN YOU AND ACIDIC BATHS AND COBALT-BEARING TOPCOATS

A trivalent line works with acidic activation and passivate baths and often **cobalt-containing topcoats/sealers.** The hazard level is much lower than a Cr(VI) line — there's no carcinogen in the main tank — but PPE is still essential.

★ **REMEMBER**

There is no part, no order, and no deadline worth your health. Scrap can be reworked. Your skin, eyes, and lungs are harder to fix. Trivalent is the safer chromate — *treat that as a reason to keep good habits, not to drop them.*

THE MASTER PPE SET (WEAR AT ALL CHEMICAL STATIONS)

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; acid finds them.
- **Face shield** — *over* the goggles for any work at the activation, passivate, or topcoat tanks.
- **Chemical-resistant gloves** — rated for the acids and the topcoat chemistry per the SDS (nitrile, neoprene, or PVC). Inspect for pinholes before each use — especially important with **cobalt-bearing** topcoats, which are skin sensitizers.
- **Chemical-resistant apron/suit** — full coverage, chest to below the knee.
- **Chemical-resistant, non-slip boots** — floors around dip tanks and rinses are wet and slick.
- **Respiratory protection** — per your shop's program/SDS. Not at the extreme Cr(VI) level, but warranted when handling **cobalt-bearing** products as powders/aerosols, or any time ventilation is degraded.

⚠ **HEADS-UP — COBALT IS A SENSITIZER**

Many trivalent passivates and topcoats contain **cobalt,** a recognized **skin and respiratory sensitizer** (and some cobalt compounds carry carcinogen classifications). Avoid skin contact, don't breathe mist/dust, wash thoroughly, and **read the SDS for every product you run** — gloves and ventilation requirements come from there.

💡 **TIP**

Put gear on *in order:* boots and apron first, then gloves, then goggles, then face shield (and respirator if required) last. Take it off in reverse, and rinse your gloves *before* removing them. This keeps contaminated gloves away from your face.

STATION	PRIMARY HAZARD	WHY IT MATTERS
01 Receive/Inspect	Mechanical; sharp/heavy parts	Cuts, pinches
02 Acid Activation	**Acid**	Chemical burns; mist irritation
03 Rinse	Residual acid; wet floor	Mild irritant; **slips**
04 Trivalent Passivate	**Acidic; possibly cobalt-bearing**	Burns/irritation; cobalt sensitization
05 Rinse	Acidic/metal-bearing water	Mild irritant; slips
06 Topcoat / Seal	**Often cobalt-bearing; chemistry per SDS**	Sensitization; irritation
07 Dry/Cure	Warm/hot air, hot parts	Burns; heat stress
08 Inspect/Hand-Off	Residual chemistry on parts	Mild irritant

⚠ **HEADS-UP**

Never eat, drink, smoke, vape, or apply cosmetics on the line. Wash thoroughly before breaks and before leaving. Keep food and drink out of the work area. Good hygiene matters with acids and cobalt, even on a "safer" line.

CHAPTER 7

EMERGENCY & FIRST-RESPONSE

WHEN SECONDS COUNT — LEARN THIS COLD, BEFORE YOU EVER NEED IT

Know these *before* you need them. **Locate, before your first shift:** the nearest **emergency eyewash** (within ~10 seconds of every chemical station), the **safety shower**, the **fire extinguisher** and alarm, the **spill kit**, and the **SDS binder.**

⚠ **HEADS-UP**

Eyewash and shower stations must always be unblocked. Never stack baskets, totes, or boxes in front of them. The day you need one is the day you can't move boxes out of the way.

• FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Passivate / acid on skin	Safety shower; flush **at least 15 minutes.** Remove contaminated clothing while flushing. Get medical attention for anything beyond minor irritation. Bring the SDS.
Passivate / acid in eyes	Eyewash; hold eyelids open and flush **at least 15 minutes**, rolling the eyes. Do not rub. Get medical attention immediately. Bring the SDS.
Topcoat (cobalt) on skin / breathed	Wash skin thoroughly; move to fresh air. Cobalt is a **sensitizer** — report repeated exposure and any rash/breathing symptoms; follow SDS first aid.
Inhaled acid mist, feel ill	Move to fresh air immediately. Report it — don't "tough it out." Seek medical attention for anything beyond brief mild irritation.
Chemical swallowed	**Do not** induce vomiting unless the SDS directs it. Follow SDS first aid; call poison control / medical help immediately.
Chemical spill	Alert others; keep people away. Clean only if trained and within your shop's "small spill" limits — otherwise evacuate and call trained spill response. Don't let spills reach a floor drain.

⚠ **HEADS-UP — LOCKOUT/TAGOUT (LOTO)**

Mandatory for *all* electrical/mechanical maintenance — heaters, pumps, hoists, dryer, ventilation. Physically lock the power off and tag it so no one can re-energize while someone works. "I'll just be a second" is how people get hurt.

⚠ **HEADS-UP — A CRITICAL ACID-MIXING RULE, FOR LIFE**

When diluting acid or making bath additions per your procedure, **always add ACID/chemical to WATER — never water to concentrated acid.** Adding water to concentrated acid releases heat so violently it can erupt acid into your face. Memory aid: *"Do as you oughta — add acid to water."*

☀ **FUN FACT**

The 15-minute flush rule isn't arbitrary — it's the time emergency-response standards found necessary to dilute and carry away most corrosive chemistry before it does deep tissue damage. That's why eyewash and shower stations must deliver a steady, hands-free flow for a full 15 minutes.

CHAPTER 8

THE SAFETY PHILOSOPHY OF A TRIVALENT CHROMATE LINE

THE SAFER, GREENER CHROMATE — BUT NOT HAZARD-FREE

People sometimes assume that because trivalent is “the RoHS-compliant safe one,” it needs no respect. That’s the wrong takeaway. **Trivalent is genuinely far safer than the old Cr(VI) line — and that’s exactly why it’s worth running it well.** The honest hazard list:

- **No carcinogen in the main tank.** Cr(III) is not a carcinogen, and there’s no Cr(VI) PEL to chase. That’s the big win — but it’s a win you keep only by *keeping the chemistry trivalent.*
- **The acids still bite.** The activation and passivate baths are acidic — burns and eye injuries are real.
- **Cobalt is a sensitizer.** Cobalt-bearing passivates and topcoats can sensitize skin and lungs; some cobalt compounds carry carcinogen classifications. Read the SDS, use gloves and ventilation.
- **Keep it trivalent.** Don’t let the chemistry oxidize to — or get cross-contaminated with — hexavalent. That protects people *and* the RoHS status of the finish.
- **Waste is easier, not absent.** No Cr(VI)-reduction step is needed, but you still must **treat the metals** (zinc, chromium, cobalt) out of the wastewater before discharge. (Part Three.)

★ **REMEMBER**

The safety philosophy of this line: **respect the acids, mind the cobalt, keep the chromium trivalent, treat the metals in the waste, and keep good hygiene.** A trivalent line isn’t “the one with no rules” — it’s the line where doing things right is *easier,* and where the biggest single rule is **don’t undo the very thing that makes it safe and compliant: its trivalent identity.**

★ **REMEMBER**

You’ve now got the whole foundation: what a conversion coating is, what makes a trivalent passivate different (passive barrier, **not** self-healing, **needs a topcoat**), why RoHS/ELV/REACH drove the switch from Cr(VI), how the eight-station line works as one connected **system,** why the order is locked, the real (lower) hazards, what to wear, what to do in an emergency, and the mindset that ties it together. Turn the page knowing the *why* — and the *how* will make a lot more sense.

PART TWO — THE LINE, STATION BY STATION (STATIONS 01–08)

STATION 01 OF 08

RECEIVE & INSPECT

"YOU CAN'T PASSIVATE ZINC THAT'S ALREADY GONE BAD."

Before any passivate chemistry touches a part, someone confirms it arrived **freshly zinc-plated and rinsed**, in the right condition, and on time. On most lines the passivate process picks up directly where the zinc line dropped off. This feels like the un-glamorous step. It isn't: a passivate is only as good as the zinc surface it's given, and **zinc ages.**

WHAT YOU'RE DOING & WHY IT MATTERS

- **Confirm the job.** Right part number, quantity, spec. Check the traveler for the **target color/type** (clear/blue, iridescent, black), the **substrate** (zinc? zinc-nickel? cadmium?), the **required topcoat,** and the **salt-spray hours** to hit.
- **Confirm freshness.** Ideally the parts come straight from plating and rinse. Parts that have sat or show **dullness or white rust** passivate poorly — flag them.
- **Inspect condition.** White rust, heavy water spotting, fingerprints, or contamination on the zinc are warning signs. Heavily corroded zinc cannot be rescued by a passivate.

OPERATOR ESSENTIALS

ITEM	GUIDANCE
Substrate check	Zinc / zinc-alloy / cadmium = good. Aluminum or bare steel = **wrong line — flag it**
Freshness	Straight from plating is best; aged/dull zinc → lean on activation, flag if white-rusted
Target color/type & topcoat	Confirm clear/blue / iridescent / black **and** the required topcoat per traveler
Salt-spray requirement	Note the hours the system must hit — it drives passivate type + topcoat choice
Handling	Clean gloves — fingerprints on zinc become defects under the passivate
Load/basket condition	Parts not nested; barrels not overpacked so solution reaches all faces

⚠ **HEADS-UP**

Mechanical hazards rule this station — sharp edges, heavy baskets, pinch points. Wear cut-resistant gloves for handling, lift with your legs, and get help for heavy loads.

• STEP-BY-STEP PROCEDURE

1. **Read the traveler.** Confirm part, quantity, substrate, target color/type, **required topcoat,** and salt-spray spec.
2. **Check freshness and condition.** Set aside anything dull, white-rusted, or contaminated for your lead.
3. **Handle with clean gloves.** Skin oils cause spotty passivation.
4. **Load for full contact and drainage.** No nesting; barrels not overpacked; every surface sees solution and every cavity drains.
5. **Flag anything unusual** (wrong substrate, aged parts, mixed lots) before the load enters activation.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Running aged or white-rusted zinc** as if it were fresh — thin color, patchy protection.
2. **Bare-handed handling** — fingerprints become passivation voids.
3. **Loading the wrong substrate** (aluminum/steel) without flagging it.
4. **Overpacking barrels / nesting parts** so solution can't reach all faces.
5. **Missing the topcoat/salt-spray callout** — and processing the wrong system.

• TEACH-BACK CHECKLIST

- [ ] Explain why **fresh** zinc passivates better than aged zinc.
- [ ] State which substrates belong on this line and what to do with aluminum/steel (flag it).
- [ ] Identify incoming conditions that need flagging (dullness, white rust, contamination).
- [ ] State where on the traveler to find the **required topcoat and salt-spray spec.**

SIGN-OFF — STATION 01

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I confirm I verified the job, substrate, target color, required topcoat, and salt-spray spec; that parts were fresh and in good condition; handled cleanly; and loaded for full contact and drainage."

STATION 02 OF 08

ACID ACTIVATION / BRIGHT DIP

“WAKE THE ZINC UP — STRIP THE SMUT, EXPOSE FRESH METAL — WITHOUT POISONING THE PASSIVATE.”

This is a quick dip in a **dilute acid** (commonly dilute **sulfuric** or a **proprietary** activator on trivalent lines — often chosen over nitric to limit oxidizer drag-in) that **dissolves the light oxide/smut film** on the zinc and a whisker of zinc itself, leaving a bright, chemically fresh, uniformly reactive surface for the passivate. It is short — usually just a **few seconds.**

• WHAT YOU’RE DOING & WHY IT MATTERS

Even fresh zinc carries an invisible oxide/smut layer that makes the passivate form unevenly — patchy color, thin spots, weak protection. The activation dip removes it and **resets the surface** so the passivate reacts the same way everywhere.

⚙ TECHNICAL STUFF — WHY OFTEN NOT NITRIC

Hex lines commonly use a dilute **nitric** bright dip. On many trivalent lines that’s avoided or minimized, because **nitric is a strong oxidizer,** and oxidizer drag-in is one of the things that can push trivalent chemistry toward forming **hexavalent** chromium. A **dilute sulfuric or proprietary** activator cleans/activates the zinc with less oxidizer risk. Concentration and time stay low on purpose — **clean and activate, don’t strip plated zinc.** Exact chemistry, concentration, and time come from the supplier TDS.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Acid	`Dilute (per TDS) — often sulfuric/proprietary, ~0.5–3%`	Avoid/minimize strong oxidizers (e.g., nitric) per TDS
Temperature	`Ambient`	No heat needed
Time	`A few seconds (per TDS)`	Short — over-dipping removes zinc
Movement	`Brief agitation/rotation`	Even contact on all faces

⚠ HEADS-UP — ACID

The activation dip is **acidic** and can cause **chemical burns** and serious eye injury, and it throws a fine mist. **PPE every time:** sealed goggles, face shield, chemical gloves, apron, boots. Know your eyewash and shower locations, and never mix chemistries.

• STEP-BY-STEP PROCEDURE

1. **Check the dip** — concentration in spec (last titration), level, and cleanliness.
2. **Confirm the prior rinse** delivered a clean part — drag-in of alkaline rinse weakens the dip.
3. **Dip for the specified short time** with brief agitation. Watch dull parts flash bright.
4. **Withdraw and drain** back to the tank.
5. **Move promptly to rinse** — an activated zinc surface re-oxidizes within minutes.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Over-dipping** — removes plated zinc and can over-etch the surface.
2. **Letting activated parts sit** before passivating — they re-oxidize and passivate unevenly.
3. **Running the dip out of spec** (too weak = no activation; too strong = over-etch).
4. **Using a strong oxidizing acid the TDS didn't call for** — risking oxidizer drag-in to the trivalent bath.
5. **Dragging alkaline rinse water in** and neutralizing the dip.

• QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Dull/smutty parts after dip	Dip too weak or spent	Titrate; refresh per TDS
Over-etched / hazy zinc	Dip too strong or time too long	Reduce time/concentration per TDS
Patchy passivate downstream	Uneven activation; parts sat too long	Re-activate; shorten transfer time

SIGN-OFF — STATION 02

Trainee: ____________________ Trainer: ____________________ Conc./Time: __________

"I confirm the activation dip was in concentration and the correct (non-oxidizer where specified) chemistry, parts were dipped the correct short time and moved promptly to rinse, and I wore required PPE."

STATION 03 OF 08

RINSE (POST-ACTIVATION)

“A RINSE ISN’T A REST STOP. IT’S A FIREWALL — AND ON TRIVALENT IT ALSO KEEPS THE CHROMIUM TRIVALENT.”

You just activated the zinc — but it’s coated in a film of activation acid. This station’s job is to **wash that acid off** before the part hits the passivate, so the activation chemistry doesn’t drift or contaminate the passivate bath. It looks like a “nothing” step. It isn’t: drag-in of acid quietly shifts the passivate’s pH, and drag-in of an **oxidizer** can even nudge the bath toward forming hexavalent chromium.

• WHAT YOU’RE DOING & WHY IT MATTERS

The activation acid clinging to the part is **drag-out** (leaving the dip) / **drag-in** (arriving at the passivate). The rinse knocks it down by dilution — *but quickly,* because activated zinc re-oxidizes if it lingers.

TIP — COUNTER-FLOW RINSING

The professional approach is a **counter-flow (counter-current) rinse**: parts move forward, water flows backward. Fresh water enters the last (cleanest) stage and overflows into the dirtier one, so the cleanest water the part sees is the last thing to touch it before the passivate. Saves water and gives the passivate a clean start.

```
FROM          +-------------+  PARTS ->  +-------------+         TO
ACTIVATE ->  |  RINSE 1    | ---------> |  RINSE 2    | ---->  PASSIVATE
             |  (dirty)    |           |  (clean)    |
             |  acid in    |           |  near-neutral|
             +-------------+           +-------------+
                  ^   overflow <----------    ^
                  +------ <- FRESH WATER FLOWS BACKWARD <- -----+
                                                      FRESH IN
```

HEADS-UP

Rinse water carries dragged-in acid and is mildly acidic. Wear **goggles, gloves, and non-slip footwear** — the floor around rinses is always wet, and **slips are the #1 injury here.**

REMEMBER

Don’t let the part dwell in the rinse. Activated zinc is reactive and re-oxidizes fast. Rinse clean, then move straight to the passivate. A long pause here undoes the activation you just did at Station 02.

• TOP MISTAKES

1. Letting the activated part sit in or after the rinse (re-oxidation).
2. A tired, contaminated rinse that drags acid/oxidizer into the passivate.
3. Treating the rinse as optional “downtime.”

SIGN-OFF — STATION 03

Trainee: ______________________ Trainer: ______________________ Date: ______________

“I confirm the rinse was clean/flowing, parts were rinsed and moved promptly to the passivate, and I wore required PPE.”

STATION 04 OF 08 · THE CONVERSION HEART

TRIVALENT PASSIVATE — THE CONVERSION FILM

COOL-TO-WARM, QUICK, HEX-FREE — THE CR(III) CONVERSION FILM FORMS HERE. BUT IT'S ONLY HALF THE SYSTEM

This is the conversion heart of the line. With a clean, freshly activated zinc surface, you dip the part in a **dilute acidic trivalent-chromium solution** for **seconds**, and the Cr(III) conversion film grows. Then you rinse and hand it to the topcoat — because on trivalent, **the passivate alone is not the finished job.**

• WHAT YOU'RE DOING & WHY IT MATTERS

The passivate solution (**Cr(III) salts + other metal salts such as cobalt + accelerators, pH ~1.8–2.5 (product-specific), cool-to-warm**) reacts with the zinc: acid dissolves a little zinc, local pH rises, and a **hydrated chromium(III)/zinc oxide-hydroxide film precipitates.** Because there's **no hexavalent chromium,** the film is a **passive barrier — no self-healing reservoir.** Color (clear/blue → iridescent → black) tracks film type/thickness and the eventual topcoat.

• OPERATOR ESSENTIALS

PARAMETER	TYPICAL RANGE	NOTES
Chemistry	Dilute **Cr(III)** salts + co-metals (e.g., cobalt) + accelerators	Per supplier TDS; **trivalent, hex-free**
pH	`~1.8–2.5 (thick-film/iridescent types often higher, ≈3–4)`	The master control variable; tightly held
Temperature	`Cool–warm (~20–60°C, product-specific)`	Trivalent often runs a bit warm to build film
Immersion time	`~30–60 s`	Short; under = thin film, over = can powder/over-etch
Agitation	`Gentle (per equipment)`	Even contact; barrels rotate slowly
Color/type	`Clear/blue / iridescent / black per job`	Final color/performance set with the topcoat

⚠ **HEADS-UP — KEEP IT TRIVALENT**

This bath is **acidic** and may be **cobalt-bearing,** so wear full PPE and keep ventilation running. The special rule here: **don't let it become hexavalent.** Don't drag oxidizers in, don't cross-contaminate with hex chemistry or shared equipment, and hold pH/temperature per TDS. Oxidation to Cr(VI) would create a carcinogen *and* break RoHS compliance.

• STEP-BY-STEP PROCEDURE

1. **Check the controls and bath.** Ventilation on; **pH and temperature in spec** (last reading); level and clarity normal.
2. **Confirm the part is freshly activated and rinsed** — and not sitting/re-oxidized.
3. **Immerse for the specified time** (~30–60 s) for the target film, with gentle agitation.
4. **Withdraw smoothly and let drag-out drain back into the tank** — recovering chemistry and keeping the line tidy.
5. **Move to the controlled rinse,** then to the topcoat. Don't let the soft film dry on the way (it'll spot).

💡 **TIP — READ THE FILM, BUT VERIFY BY SPEC**

A thin trivalent gives a faint blue/clear; thicker types show more iridescence. Color is a *rough* in-process cue, **but trivalent color does not signal protection the way the old hex yellow did.** The real verdict is the topcoated system's **salt spray.** Pull at the spec'd time/film, then rely on the topcoat + lab data.

⚠ **HEADS-UP — THE FRESH FILM IS SOFT**

What you just grew is a soft, hydrated film, not a hard coating. It's easily wiped or marked until it's topcoated and cured. Handle gently; don't stack, rub, or let parts bang together before the topcoat.

⚙ **TECHNICAL STUFF — WHY CONTROL IS EVERYTHING**

Trivalent baths are **more sensitive than hex** to pH, temperature, metal balance, and contamination. Too short/cold/dilute → thin film, low corrosion. Too long/aggressive → over-etch and a loose film. Drift the metal balance or let in contaminants → inconsistent color and salt spray. Bath control (by your lab, per TDS) is what makes trivalent repeatable — there's no self-healing reservoir to cover for a sloppy bath.

TOP MISTAKES NEW OPERATORS MAKE

1. **Treating the passivate as the whole finish** — forgetting it needs the topcoat to perform.
2. **Letting the bath drift** (pH/temperature/metal balance) — trivalent is unforgiving of this.
3. **Allowing oxidizer drag-in or hex cross-contamination** — risking Cr(VI) formation.
4. **Letting the soft film dry or get rubbed** before rinse/topcoat.
5. **Under- or over-immersing** — thin film or over-etched, loose film.

QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
No/weak film, thin color	Dip too short/weak; cold; out of pH; stale/poorly activated zinc	Adjust time; check pH/temp/chemistry; re-activate
Powdery / loose film	Over-immersion; bath too aggressive; pH off	Shorten dip; verify pH/temp per TDS
Patchy / uneven film	Poor activation; nesting; uneven agitation	Re-activate; reload; check agitation
Inconsistent color run-to-run	Bath drifting (pH/metal balance/temp)	Titrate/adjust per TDS (lab)

⚠ **HEADS-UP — THE COMPLIANCE CHECK THAT'S ALSO A QUALITY CHECK**

If a part ever tests positive for **Cr(VI)**, stop — that's both a RoHS failure and a sign the bath has been oxidized or cross-contaminated. Investigate the activation chemistry, shared equipment, and bath condition before running more. Keeping the bath trivalent is a daily habit, not a one-time setup.

• TEACH-BACK CHECKLIST

- [ ] State that the bath is **trivalent (Cr III), hex-free,** acidic, and possibly **cobalt-bearing** — and name the controls/PPE.
- [ ] Explain why the passivate is **not self-healing** and **needs a topcoat** to hit spec.
- [ ] Explain why trivalent color does **not** read protection the way hex yellow did.
- [ ] Describe how to **keep the bath trivalent** (no oxidizer drag-in, no hex cross-contamination, hold pH/temp).
- [ ] Describe correct **drag-out** handling.

★ **REMEMBER**

This station does two jobs that the old hex tank rolled into one: it **converts the surface** (the passivate film) but it does **not** deliver the final corrosion numbers on its own. Carry that forward to Station 06 — the passivate you just grew is waiting for its topcoat partner.

⚠ **HEADS-UP — SAFETY GATE**

No operator runs Station 04 solo until the **safety / PPE / SDS** and **keep-it-trivalent** rows of the Training Matrix are signed. It's a far safer tank than the old hex one — but it's still acidic, possibly cobalt-bearing, and must be kept hex-free.

SIGN-OFF — STATION 04

Trainee: ____________________ Trainer: ____________________ pH/Temp: ________ Color/film: ________

"I confirm the bath was in pH/temperature spec, I passivated for the correct time, kept the chemistry trivalent (no oxidizer drag-in / no hex cross-contamination), handled the soft film and drag-out correctly, and wore required PPE."

STATION 04 · THE CENTRAL OPERATOR IDEA

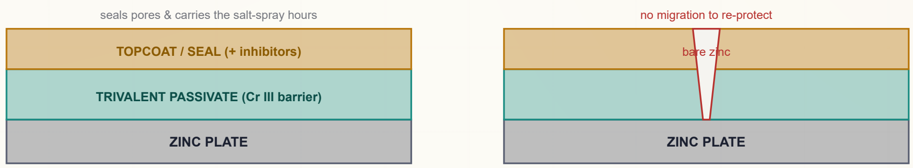
WHY THERE'S NO SELF-HEALING

PASSIVE BARRIER, NOT ACTIVE REPAIR — AND HOW THE PASSIVATE + TOPCOAT SYSTEM MAKES UP FOR IT

The old hex chromate could **patch its own scratches** because it held a reservoir of mobile **Cr^{6+}** that leached out and re-passivated bare zinc. **Trivalent has no Cr^{6+} — so it cannot do that.** A trivalent film is a **passive barrier:** it protects by *covering,* not by *repairing*. So how does a trivalent finish reach 120–500+ hours of salt spray? By being a **system.**

PASSIVE BARRIER SYSTEM — STACKED & SEALED LAYERS (NO SELF-REPAIR)

1 · THE LAYERED SYSTEM | **2 · A SCRATCH**



Protection by covering & sealing — no mobile Cr^{6+}

Damage stays damaged — coverage & topcoat must prevent it

Trivalent wins by stacking & sealing good layers — not by self-healing. The topcoat is structural.

★ **REMEMBER**

This is the one picture to carry from the whole manual: **trivalent wins by stacking and sealing good layers, not by self-healing.** That's why the topcoat is (almost) never optional, why coverage and bath control matter so much, and why a scratch in service is more serious on trivalent than it was on hex. **The passivate + topcoat is the finish.**

☀ **FUN FACT**

Because trivalent can't re-heal a scratch, designers and platers lean harder on **getting full, even coverage the first time** and on **robust topcoats.** In a way, trivalent made the *whole process* more disciplined: with no self-healing safety net, every layer has to be right. Safer chemistry, tighter craft.

STATION 05 OF 08

RINSE (CONTROLLED)

"WASH OFF THE DRAG-OUT — AND HAND A CLEAN FILM TO THE TOPCOAT."

The part leaves the passivate wet with acidic, possibly cobalt-bearing drag-out that must come off before the topcoat — but the **fresh film is soft.** So: **controlled, gentle rinse** — clean enough to prep for the topcoat, gentle enough not to damage the film.

• WHAT YOU'RE DOING & WHY IT MATTERS

- **Remove drag-out** so you don't contaminate the topcoat tank or ship dirty parts.
- **Preserve the soft film** — gentle handling, no scrubbing.
- **Set up the topcoat** — a clean, evenly rinsed film lets the topcoat wet out and bond uniformly.

⚠ **HEADS-UP**

Unlike the old hex line, you're **not** trying to protect a leachable Cr(VI) reservoir here (there isn't one) — but you *are* protecting a soft film and prepping it for the topcoat. Follow the TDS on rinse temperature; some trivalent systems are fine with cool, others tolerate mild warmth — **the TDS, not habit, decides.**

★ **REMEMBER**

Clean and gentle, then straight to the topcoat. A good rinse here is what lets the topcoat — the performance step — do its job. Contaminate the topcoat tank with drag-out and you degrade every part that follows.

⚠ **HEADS-UP**

This rinse water carries acid and dissolved metals (zinc, chromium, possibly cobalt). It goes to **waste treatment** (metals removal), not down a drain. Wear goggles, gloves, and non-slip footwear.

• TOP MISTAKES

1. **Skimping** — leaves drag-out that contaminates the topcoat tank.
2. **Rough handling** of the still-soft film.
3. **Ignoring the TDS** on rinse conditions.

SIGN-OFF — STATION 05

Trainee: ____________________ Trainer: ____________________ Date: ____________

"I confirm I rinsed per TDS to remove drag-out without damaging the soft film, handled it gently, routed rinse water to waste treatment, and prepped a clean surface for the topcoat."

STATION 06 OF 08 · THE PERFORMANCE STEP

TOPCOAT / SEAL — THE PERFORMANCE STEP

"ON TRIVALENT, THIS IS WHERE YOU ACTUALLY MEET THE SPEC — IT'S NOT OPTIONAL POLISH."

This is the station the old hex manual treated as optional. **On trivalent it is the star.** Most trivalent finishes reach their required salt-spray hours **because of the topcoat/seal,** not the passivate alone. Whether you run one — and which one — comes from the **job spec and TDS,** and on most trivalent jobs the answer is *yes, run it.*

• WHAT YOU'RE DOING & WHY IT MATTERS

A topcoat/seal is a thin additional layer applied over the rinsed passivate that **closes pores and microcracks, adds a barrier, carries corrosion inhibitors,** and often sets **color** and **lubricity (torque-tension control)** on fasteners. On trivalent it's usually doing **load-bearing corrosion work** to make up for the missing self-healing.

TOPCOAT/SEAL TYPE	WHAT IT ADDS	CR(VI)?	NOTE
Silica / inorganic (nano-silica) seal	Barrier, big salt-spray boost	No	Very common on trivalent
Organic / polymer topcoat	Corrosion life, color, **torque-tension control**	No	Common for fasteners and color
Cobalt-containing seals/topcoats	Higher corrosion performance	No (Cr-free)	**Cobalt = sensitizer; check SDS / PPE**

⚠ **HEADS-UP — COBALT AND CHEMISTRY HAZARDS**

Many high-performance trivalent topcoats/sealers contain **cobalt** (a **sensitizer**) or other reactive chemistry. These are **Cr-free** (good — they keep the finish hex-free), but they have their **own** hazards. **Read the SDS,** use the specified gloves and ventilation, and avoid skin/respiratory contact.

★ **REMEMBER**

On **hex,** the seal was optional polish on top of a self-healing film. On **trivalent,** the topcoat is usually **mandatory** — it's how the *passive* system reaches the corrosion numbers a customer specifies. If a traveler calls for a salt-spray spec, assume the topcoat is part of how you'll hit it, and **never skip it to save a step.**

OPERATOR ESSENTIALS

PARAMETER	GUIDANCE
Run it at all?	On trivalent, **usually yes** — per spec/traveler; it's how you meet salt spray
Type	Per TDS — silica, organic/polymer, or cobalt-bearing (all Cr-free)
Concentration / temp / time	Per product (some warm dips; many are dried/cured in place)
Cure	Often a **warm cure** required (Station 07) — follow TDS

STEP-BY-STEP PROCEDURE

1. **Check the traveler** — which topcoat, and the target salt-spray.
2. **Confirm the passivate film is intact** and properly rinsed first.
3. **Apply per TDS** — correct concentration, temperature, dwell.
4. **Drain** and move to dry/cure. Many topcoats are **dried/cured in place** (no rinse after).
5. **Handle gently** — the stack (passivate + topcoat) is still curing.

TOP MISTAKES

1. **Skipping or shortchanging the topcoat** — the #1 cause of failed salt spray on trivalent.
2. **Treating a cobalt topcoat as harmless** — it's Cr-free but a sensitizer.
3. **Topcoating over a damaged/contaminated passivate** — you lock in the defect.
4. **Wrong cure** (Station 07) — an uncured topcoat under-performs.

TEACH-BACK CHECKLIST

- [ ] Explain why the topcoat is **usually mandatory** on trivalent (delivers the corrosion performance).
- [ ] Name the topcoat families and which carry **cobalt** (sensitizer) hazards.
- [ ] Explain why topcoating over a damaged passivate is a mistake.
- [ ] State that topcoats are **Cr-free** (they keep the finish hex-free) but have their own hazards.

SIGN-OFF — STATION 06

Trainee: ____________________ Trainer: ____________________ Topcoat type: __________

"I confirm I applied the specified topcoat/seal per TDS over an intact, clean passivate, used the correct PPE for cobalt/chemistry hazards, and understand the topcoat is how the system meets its salt-spray spec."

STATION 07 OF 08 · THE MAKE-OR-BREAK STEP

DRY-OFF & CURE

"CURE IT RIGHT PER THE TDS — WARM IS OFTEN REQUIRED HERE, BUT NEVER OVER THE LIMIT."

This is where the trivalent line *differs* from the hex line's "always dry cool" rule. Many trivalent topcoats and thick-film passivates **require a warm cure** to cross-link/consolidate and deliver their salt-spray performance. But there's still a **TDS maximum** — go too hot and you can **craze (microcrack) the film** and *lose* corrosion resistance. So the rule here is **cure per the TDS — warm if specified, never over the limit.**

• WHAT YOU'RE DOING & WHY IT MATTERS

Drying removes surface water; **curing** sets the passivate/topcoat into a coherent, durable system so the part can be handled, packed, and (if needed) painted. The catch: **the right temperature is product-specific.** Too cool/short and a topcoat won't fully cure (under-performs); too hot and you craze the film (also under-performs — often invisibly, then it fails salt spray).

PARAMETER	GUIDANCE
Temperature	**Per TDS** — many trivalent topcoats specify a **warm cure** (commonly up to ~80–100°C, product-specific); **stay under the TDS maximum**
Method	Warm-air oven, spin-dry + warm cure, per the product
Time	Long enough to fully cure; not so hot/long that it crazes
What NOT to do	Don't exceed the TDS temperature limit; don't skip a required cure

⚠ **HEADS-UP — THE CURE RULE OF THIS LINE**

Cure per the TDS. Unlike the old hex "dry cool only" rule, trivalent topcoats often *need* warmth to perform — but **over-baking past the TDS limit crazes the film and destroys corrosion resistance,** often invisibly (looks fine, fails salt spray). Don't carry the hex habit ("never use heat") *or* the phosphate habit ("crank the oven") — run *this* product by *its* number.

⚙ **TECHNICAL STUFF**

A fresh passivate is a hydrated Cr(III)/zinc oxide-hydroxide film; the topcoat is a sealer/polymer that **cures** with the right heat/time. Proper cure consolidates the system. **Flash-overheating** shrinks and **crazes** the film (microcracks), opening paths for corrosion — and unlike hex there's no self-healing to compensate. The damage is usually invisible, which is why the cure window is a **hard rule from the TDS,** not a guess.

★ **REMEMBER**

The system keeps hardening as it cures. Even after a proper cure, handle parts gently for the first several hours to ~24 h and avoid heavy stacking/abrasion. Don't pack hot or fresh parts tightly.

SIGN-OFF — STATION 07

Trainee: ____________________ Trainer: ____________________ Cure temp/time: __________

"I confirm I dried/cured within the TDS window (warm where specified, not over the limit), did not craze the film, and handled the curing system gently."

STATION 08 OF 08 · THE FINISH LINE

INSPECT & HAND-OFF

"LAST CHANCE TO CATCH IT BEFORE IT SHIPS."

A final check confirms the finish is the **right color/type, uniform and fully covering,** with an **intact, cured topcoat,** then the parts are packed or sent on (to paint, assembly, or shipping).

• WHAT YOU'RE CHECKING

CHECK	WHAT IT TELLS YOU	HOW
Color / type	Right finish	Visual vs. standard (clear/blue / iridescent / black)
Coverage / uniformity	No bare or thin spots	Visual, all faces
Topcoat integrity	Topcoat applied, even, cured (no streaks/blush/missing areas)	Visual; light touch
Adhesion (no rub-off)	Film/topcoat cured and adherent	Light wipe — finish should not rub off
No white rust	Zinc still protected	Visual
Salt spray (per spec)	Actual corrosion life of the *system*	ASTM B117 on samples (lab)

💡 **TIP**
The fast shop checks are **color/coverage** and a **light touch** for an intact, cured topcoat. But remember — **trivalent color doesn't read protection the way hex yellow did.** The real verdict is **salt spray (ASTM B117)** on the topcoated system. If parts look fine but fail salt spray, suspect a **skipped/weak topcoat or a bad cure.**

★ **REMEMBER**
The system is **still curing.** Don't judge final hardness the instant it leaves the oven, and don't pack tightly while parts are warm/fresh. Give the topcoat its cure time before rough handling.

⚠ **HEADS-UP**
Even finished, parts may carry trace chemistry (including cobalt from topcoats). Maintain hygiene — **no eating/drinking at inspection,** wash hands, and follow your shop's handling rules.

• TEACH-BACK CHECKLIST

- ☐ Identify the trivalent colors (clear/blue, iridescent, black) and that they don't read protection like hex yellow.
- ☐ Demonstrate the light-rub check and say what rub-off means.
- ☐ Explain why **salt spray on the topcoated system** is the real verdict.
- ☐ Explain why the system keeps curing and how that affects handling/packing.

SIGN-OFF — STATION 08

Trainee: ____________________ Trainer: ____________________ Date: ____________

"I confirm I verified color/type, full coverage, an intact cured topcoat, checked for white rust, understood salt spray is the real verdict, and handled/packed the curing parts correctly."

PART THREE — ACROSS THE LINE
READING AND CONTROLLING THE FILM

COLOR & COATING CONTROL

COLOR TELLS YOU FILM TYPE — BUT ON TRIVALENT, SALT SPRAY (ON THE TOPCOATED SYSTEM) TELLS YOU PROTECTION

Because the film is so thin, you "read" a trivalent passivate by **color and coverage,** backed by lab checks. But the key change from hex: **color comes from film type + topcoat — not from a leachable chromate — so color is a weaker protection signal than it was on hex.** Judge protection by **salt spray on the complete system.**

COLOR / TYPE	RELATIVE FILM	TYPICAL SALT-SPRAY TO WHITE RUST, TOPCOATED SYSTEM (APPROX., SPEC-DEPENDENT)
Clear / blue-bright (thin trivalent + topcoat)	Lightest	`~72–150+ hr`
Iridescent (thicker trivalent + topcoat)	Medium–heavy	`~150–300+ hr`
Black (thick trivalent + black topcoat)	Medium–heavy	`varies; often 120–500+ hr by system`
Thick-film trivalent + high-performance topcoat	Heaviest	up to **500+ hr** in robust systems

⚠ HEADS-UP — VERIFY THE NUMBERS

The salt-spray hours above are **ballpark ranges for illustration only,** and they assume a **topcoated system.** Actual performance depends on zinc thickness (B633 Class), the specific passivate, the topcoat, the cure, and test detail. **Always qualify to the customer's spec and your own salt-spray data — never quote these numbers as a guarantee.** A bare trivalent passivate *without* a topcoat typically gives far fewer hours.

★ REMEMBER

On trivalent, **don't chase a shade to judge protection.** Match the spec'd color/type, run the required topcoat, and **prove performance with salt spray.** A pretty part that skipped or under-cured its topcoat will fail.

⚙ TECHNICAL STUFF

Levers that move trivalent film/color: **dip time, bath concentration, pH, temperature, zinc condition** — and then the **topcoat** on top. Thick-film passivates (run warmer, sometimes higher pH) build heavier films and more iridescence. But the biggest lever on *corrosion numbers* is the **topcoat + proper cure,** because that's what seals the passive barrier. Bath control (by your lab) keeps it all repeatable.

THE ONE RULE UNIQUE TO TRIVALENT

DRYING & CURING THE SYSTEM

THE TRIVALENT TWIST: CURE PER THE TDS — WARM IS OFTEN REQUIRED, OVER-BAKING STILL KILLS IT

This deserves its own page because the trivalent rule is **different from hex** and easy to get wrong. The fresh passivate + topcoat is a system that must **cure,** and the cure conditions are **product-specific:**

1. **Drying** — remove surface water.
2. **Curing** — set the topcoat (and consolidate the passivate). **Many trivalent topcoats require a *warm* cure** to perform — this is the opposite of the hex “dry cool only” rule.

But there’s a ceiling: **over-baking past the TDS limit flash-dehydrates and crazes (microcracks) the film,** opening corrosion paths — and with no self-healing to compensate, the part just fails. The killer is that the damage is usually **invisible:** the part looks fine and fails salt spray.

DO

- Cure **per the TDS** (warm where specified).
- Use the specified oven/method and time.
- Let parts age/cure several hours to ~24 h before rough handling.
- Trust **salt spray** on the topcoated system as the verdict.

DON’T

- **Skip a required cure** (uncured topcoat under-performs).
- **Over-bake** past the TDS limit (crazes the film).
- Assume the hex “never use heat” rule applies here — it often doesn’t.
- Assume “looks dry, looks fine” = good.

★

REMEMBER

Cure per the TDS: warm if the product calls for it, never over the limit. If you remember one *process* rule unique to trivalent, make it this — the cure is not optional and not “whatever the oven’s set to.”

☀

FUN FACT

This is a great example of why you run *each* chemistry by *its own* TDS: the hex chromate manual says “never dry hot,” the iron-phosphate manual says “a hot oven is mandatory,” and the trivalent manual says “use the *specific* warm cure on the data sheet.” Same shop, three different rules — because three different chemistries.

WHY TRIVALENT ISN'T SELF-HEALING

THE PASSIVATE + TOPCOAT SYSTEM

WHY TRIVALENT ISN'T SELF-HEALING — AND HOW A LAYERED SYSTEM MAKES UP FOR IT

The mechanism, step by step:

1. The trivalent passivate forms a **hydrated Cr(III)/zinc oxide-hydroxide barrier film** with **no hexavalent chromium** in it.
2. Because there's no mobile Cr^{6+}, the film **cannot leach out to re-protect a scratch** — it is **passive,** protecting by *covering,* not *repairing.*
3. To reach hex-level corrosion performance, the system adds a **topcoat/seal** that **closes pores/microcracks, adds a barrier, and carries corrosion inhibitors** (often cobalt and/or silica chemistry).
4. With **good coverage, a robust topcoat, and a proper cure,** the *system* delivers the salt-spray numbers — without any carcinogen and without self-healing.

This is **passive (barrier) protection** delivered as a **system,** versus hex's **active (self-healing) protection** delivered by one film.

★ **REMEMBER**

Passive system vs. active film. Hex healed itself with mobile Cr^{6+} (the magic *and* the carcinogen). Trivalent has no Cr^{6+} — so it wins by **stacking and sealing layers** and **controlling the process.** The topcoat isn't a bonus; it's a structural part of the protection.

⚙ **TECHNICAL STUFF**

Because there's no self-healing safety net, two things matter more on trivalent than they did on hex: **(1) full, even coverage the first time** (a missed or thin spot stays a weak spot), and **(2) topcoat integrity and cure** (the sealed barrier is doing the corrosion work). This is why trivalent lines obsess over bath control, coverage, topcoat, and cure — the system has to be right, because it can't fix itself later.

⚠ **HEADS-UP**

In service, a **deep scratch through to bare zinc is more serious on trivalent** than it was on hex, because nothing migrates to re-protect it. Design, handling, and assembly should respect that — and parts should be inspected for coating damage before/after assembly where corrosion is critical.

THE DECISION DRIVING THE INDUSTRY

HEX VS. TRIVALENT

WHY THE WORLD MOVED TO CR(III) — AND WHAT YOU TRADE FOR COMPLIANCE

FACTOR	TRIVALENT (CR III) — THIS MANUAL	HEXAVALENT (CR VI) — PREVIOUS MANUAL
Toxicity	**Far less hazardous** (not a carcinogen)	**Carcinogen**
Self-healing	**No** — passive barrier; **needs topcoat/seal**	**Strong** (mobile Cr^{6+})
Corrosion performance	**Excellent as a system** (passivate + topcoat)	Excellent on its own, cheaply
Cost / steps	More (topcoat usually required)	Less (bare dip can suffice)
Color range	Clear/blue and black common; iridescent; **not** deep hex yellow	Clear, yellow, olive, black (classic)
RoHS / ELV / REACH	**Compliant — the modern standard**	**Restricted**
Where used	**Default** for new commercial/automotive/electronics/fasteners	Phasing out; survives in some defense/industrial

• THE REGULATORY DECISION TREE (PLAIN VERSION)

- **Is the part for electronics (RoHS), automotive (ELV), or an EU/REACH-governed market? → Trivalent (or other Cr-free).** This is the normal case, and trivalent is the answer.
- **Is it a legacy application that still permits hex (some defense/industrial)?** → Hex *may* still be used there with full Cr(VI) controls — but those exemptions narrow over time, and **trivalent is the default everywhere else.**
- **When in doubt?** → Specify **trivalent** and **confirm the customer spec and current regulations.**

★ **REMEMBER**

The honest summary: **trivalent is the RoHS-compliant, far-safer modern standard — a passive-barrier system (passivate + topcoat) that you choose specifically because it's hex-free.** You trade hex's self-healing and bare-dip cheapness for **compliance, safety, and a topcoat step.** For almost all new commercial work, that's the right trade — and the one the law requires.

⚠ **HEADS-UP**

Choosing trivalent for compliance only *stays* compliant if you **keep it trivalent** — no oxidation to Cr(VI), no cross-contamination with hex chemistry/equipment. The compliance lives in the chemistry; protect it.

EASIER THAN HEX — BUT STILL TREAT THE METALS

WASTE TREATMENT

MUCH SIMPLER THAN HEX — NO CR(VI)-REDUCTION STEP — BUT YOU STILL TREAT THE METALS

Here's a genuine perk of trivalent: **waste treatment is much easier than on a hex line.** Because there's **no hexavalent chromium,** you **don't need the $Cr^{6+} \rightarrow Cr^{3+}$ reduction step** that defines hex waste treatment. But the wastewater still carries **dissolved metals** — zinc, chromium (as Cr III), and often **cobalt** — and those must be removed before discharge.

The typical approach:

1. **pH adjustment / precipitation.** The metal-bearing wastewater (rinses, dumped baths, drag-out) is **pH-adjusted (neutralized/raised)** so the dissolved metals **precipitate as metal hydroxides** (chromium, zinc, cobalt), which are settled/filtered out as sludge for proper disposal.
2. **Test and discharge.** Treated water is tested and discharged only within permit limits. The sludge is managed as regulated waste.

★ **REMEMBER**

No Cr(VI) reduction needed — but still treat the metals. Trivalent waste is "raise pH, precipitate the metals, remove the sludge." Simpler and safer than hex's reduce-then-precipitate, **but metals (incl. cobalt) still never go down a drain untreated.**

⚠ **HEADS-UP — KEEP IT CR(VI)-FREE ON THE WASTE SIDE TOO**

Don't introduce strong oxidizers into trivalent waste streams, and don't mix trivalent and hex wastes carelessly. The point of running trivalent is a hex-free operation — protect that all the way through to the drain.

⚙ **TECHNICAL STUFF**

Cobalt and chromium hydroxide precipitation is pH-dependent, so a trivalent waste-treatment system is essentially a controlled pH-and-precipitation process. Because cobalt has its own discharge limits, many shops monitor it specifically. Your facility's permit and treatment SOP define the exact targets — follow them.

☀ **FUN FACT**

On a hex line, the dramatic yellow→green color change of $Cr^{6+} \rightarrow Cr^{3+}$ reduction is the headline of waste treatment. On a trivalent line that whole step simply **doesn't exist** — the chromium is *already* trivalent. One less hazardous chemical to dose, one less thing to get wrong. That's the "greener chromate" showing up in the back of the shop, not just on the part.

THE ENGINE OF CORROSION PERFORMANCE

THE TOPCOAT / SEAL SYSTEM

ON TRIVALENT, THE TOPCOAT ISN'T A BONUS — IT'S THE ENGINE OF THE CORROSION PERFORMANCE

A **topcoat/seal** is a thin layer over the rinsed passivate that **closes pores/microcracks, adds a barrier and corrosion inhibitors, sets color, and controls lubricity.** On trivalent it's usually **required** to meet the spec.

TYPE	ADDS	CR(VI)?	NOTES
Silica / inorganic (nano-silica) seal	Barrier, large salt-spray boost	No	Very common on trivalent
Organic / polymer topcoat	Corrosion life, color, **torque-tension control**	No	Common for fasteners and color
Cobalt-containing seal/topcoat	Higher corrosion performance	No (Cr-free)	**Cobalt = sensitizer; SDS/PPE**

• WHY FASTENERS CARE ABOUT TOPCOATS

A topcoat sets the **coefficient of friction,** which controls the **torque-tension** relationship when bolts are tightened. ASTM **F1941** and automotive OEM fastener specs often require a specified friction range — and on trivalent the topcoat both **delivers corrosion hours** and **dials in torque-tension.** Two jobs, one layer.

★ **REMEMBER**

On **trivalent,** the topcoat is doing **load-bearing corrosion work** to make up for the missing self-healing, *plus* (on fasteners) setting torque-tension. **Skipping or under-curing it is the fastest way to fail a trivalent job.**

⚠ **HEADS-UP**

Topcoats are **Cr-free** (they keep the finish hex-free — good), but many contain **cobalt** or other reactive chemistry with their **own** hazards. Know which topcoat your shop runs and apply the matching SDS controls.

⚙ **TECHNICAL STUFF**

On hex, the self-healing film already performed, so a seal was a bonus. On trivalent, the seal/topcoat is structural: it seals the slightly porous passive barrier and adds inhibitor chemistry — which is precisely why trivalent salt-spray data is almost always quoted for the **topcoated system,** not the bare passivate.

HOW WE PROVE IT WORKED

QUALITY CHECKS

A TRIVALENT FINISH IS JUDGED BY CORROSION LIFE OF THE *SYSTEM* AND FILM/TOPCOAT INTEGRITY — NOT BY CHASING A SHADE

CHECK	WHAT IT TELLS YOU	HOW
Color / type	Right finish (weak protection signal on trivalent)	Visual vs. standard (clear/blue / iridescent / black)
Coverage / uniformity	No bare/thin spots	Visual, all faces
Topcoat integrity / no rub-off	Topcoat applied, cured, adherent	Light wipe — finish must not rub off
Salt spray (white rust)	Corrosion life of the topcoated **system**	**ASTM B117** salt-fog; hours to white rust per spec
Adhesion (if painted)	Paint grips the system	**ASTM D3359** crosshatch/tape on painted samples
Torque-tension (fasteners)	Friction in spec	Per F1941 / OEM method
Hex-free verification (as required)	RoHS compliance	Spot/lab test for Cr(VI) per customer/spec

★

REMEMBER

The real verdict is **salt spray (ASTM B117)** on the **topcoated system** — hours to **white rust** (zinc) then **red rust** (steel) per the customer spec. Color and coverage are fast in-process cues, but on trivalent **color does not equal protection.** If salt spray fails but the part looked fine, suspect a **skipped/weak topcoat or a bad cure.**

⚙

TECHNICAL STUFF — WHAT “SALT SPRAY” IS

Samples sit in a sealed cabinet sprayed with salt fog (ASTM B117); you record the *hours* until white rust (zinc) or red rust (steel). More hours = better protection. Customers specify and verify corrosion requirements this way — and it’s how you’d qualify a passivate/topcoat/cure combination.

💡

TIP

Keep retained samples and a **color standard** at inspection, and run periodic salt spray on production samples so you catch a drifting passivate bath, a weak topcoat tank, or an out-of-window cure *before* a customer does. For RoHS-sensitive customers, keep your **hex-free** documentation/test records current.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

SYMPTOM	LIKELY CAUSE(S)	FIRST THINGS TO CHECK
Low / failing salt spray (looks fine, fails test)	**Topcoat skipped/weak/under-cured;** thin passivate; bad cure	**Verify topcoat applied & cured per TDS;** check passivate film/bath; check cure temp/time
No / weak / thin color	Passivate dip too short/weak; cold; out of pH; stale/poorly activated zinc	Adjust time; check pH/temp/chemistry; re-activate; confirm zinc freshness
Topcoat defects (streaks, blush, missed areas, tacky)	Contaminated topcoat tank; poor rinse before topcoat; wrong/under-cure	Clean/maintain topcoat per TDS; improve Station 05 rinse; fix cure
Powdery / loose passivate	Over-immersion; bath too aggressive; pH off	Shorten dip; verify pH/temp per TDS
Crazed / microcracked film	**Over-baked cure** above TDS limit	**Drop cure temp to TDS window;** verify oven calibration
Patchy / uneven color	Poor activation; nesting/overpacked barrels; uneven agitation	Re-activate; reload; check agitation
White rust appearing	Thin passivate; **missing/weak topcoat;** damaged film; humid storage	Run correct type **+ topcoat**; protect cured parts; check cure
Color won't repeat run-to-run	Bath drifting (pH/metal balance/temp); zinc condition varies	Titrate/adjust per TDS (lab); standardize incoming zinc
Part tests positive for Cr(VI) / RoHS fail	Oxidizer drag-in; hex cross-contamination; bath/process upset	**Stop; investigate oxidation/contamination source;** check activation chemistry, shared equipment, bath condition

💡 **TIP**

The fastest diagnostic instinct on trivalent: **failing salt spray? suspect the topcoat and the cure first** (that's where the performance lives), then the passivate. No color? Suspect activation and the passivate dip. Crazed? Over-baked cure. RoHS fail? Hunt for an oxidizer or hex cross-contamination. The defect usually names the stage that failed it.

⚠ **HEADS-UP**

Before you change bath chemistry, pH, temperature, topcoat, or cure settings, confirm with your lead and the supplier TDS. Trivalent baths are **sensitive** — and any change that risks oxidizing the chromium toward Cr(VI) is a compliance issue, not just a quality one.

YOUR DECODER RING

GLOSSARY — EVERY TERM, PLAIN AND SIMPLE

ANYTIME A WORD IN THIS MANUAL MADE YOU PAUSE, FLIP BACK HERE

Activation / bright dip: A quick dilute-acid dip (often sulfuric/proprietary on trivalent) that strips oxide/smut off zinc and re-exposes fresh, reactive metal before passivating.
Accelerator / co-metal: Additives (cobalt and other metal salts, fluorides, etc.) that build the trivalent film and boost corrosion resistance. Per TDS.
Ambient: Room temperature; no heater.
ASTM B117: The salt-spray (salt-fog) corrosion test; results in *hours* to white/red rust.
ASTM B633: Spec for electrodeposited zinc on steel, including supplementary finish **Types** and thickness **Classes**; current revisions include **trivalent** options.
ASTM F1941: Spec for electrodeposited coatings on threaded fasteners (often with torque-tension/topcoat requirements).
Black trivalent: A thicker-film trivalent finish, usually with a black topcoat, giving a deep black appearance.
Cadmium: A plated metal historically chromated like zinc; now restricted in many uses.
Class (B633): The *zinc thickness* / service condition (separate from Type).
Cobalt: A metal commonly used in trivalent passivates/topcoats to boost corrosion resistance; a **skin/respiratory sensitizer** with some compounds carrying carcinogen classifications — check the SDS.
Conversion coating: A film formed by chemically reacting (converting) the metal's own surface — no electric current.
Cr(III) / trivalent chromium: Chromium in the +3 state; far less hazardous (not a carcinogen); the basis of RoHS-compliant passivates — this manual.
Cr(VI) / hexavalent chromium: Chromium in the +6 state; a confirmed human carcinogen; the basis of the old classic chromates — **restricted by RoHS/ELV/REACH, and never to be formed on this line.**
Drag-out / drag-in: Solution clinging to a part leaving a tank (drag-out) and contaminating the next (drag-in) — on trivalent, control it to protect bath balance and (with oxidizers) the trivalent identity.
ELV: End-of-Life Vehicles directive — restricts Cr(VI) in automotive parts.
Iridescence: The faint rainbow/blue shimmer of a thin film (interference); on trivalent it reads film thickness, not chromium content.
Passivate / passivation: The conversion coating formed on plated zinc by an acidic chromium solution; on trivalent it's a **passive barrier** film (not self-healing).
Passive vs. active protection: Passive = a barrier that protects by covering (trivalent). Active = a film that repairs damage (hex self-healing).
pH: Acidity/alkalinity scale (0 acid – 14 base; 7 neutral). Trivalent passivate baths run ~1.8–2.5 (thick-film types often higher).
Precipitation (waste): Raising pH so dissolved metals (Cr III, zinc, cobalt) drop out as hydroxide sludge for disposal. Trivalent needs **no Cr(VI)-reduction step.**
REACH: EU regulation placing hexavalent-chromium substances under authorization/restriction.
RoHS: Restriction of Hazardous Substances — restricts Cr(VI) in electrical/electronic equipment; the main driver for trivalent.
Salt spray (B117): The corrosion test that is the real verdict on a trivalent **system** (passivate + topcoat).
Self-healing: The hex-chromate ability to re-protect a scratch via mobile Cr^{6+} — **trivalent does not have this.**
Substrate: The base surface being coated — here, electroplated zinc / zinc alloy / cadmium.
TDS / SDS: Technical Data Sheet (how to run a product) / Safety Data Sheet (its hazards). Follow both.
Thick-film trivalent: A trivalent passivate run to build a heavier film (often warmer, sometimes higher pH) for more corrosion resistance and iridescence.
Topcoat / seal: A thin layer over the passivate (silica, organic/polymer, often cobalt) that seals pores, adds inhibitors/barrier, color, and lubricity — **usually mandatory on trivalent.**
Type (B633): The supplementary finish on the zinc; verify exact (trivalent) designations against the current revision.
White rust: Powdery white zinc-corrosion product; the first failure a passivate + topcoat is meant to delay.
Zinc / zinc-alloy plating: The electrodeposited metal layer the passivate converts (zinc, zinc-nickel, zinc-iron, zinc-cobalt).

TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

★ **REMEMBER**

An operator is "line-qualified" only when every station they run is signed. **Nobody runs the passivate or topcoat tanks, or handles waste, solo until the safety/PPE/SDS row is signed.** Safety qualification comes first, always.

Sign-off levels: O = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	**Safety / PPE & SDS, incl. cobalt & acid hazards (all stations)**	____	________	________	______
2	Station 01 — Receive & Inspect (freshness, substrate, topcoat/spec)	____	________	________	______
3	Station 02 — Acid Activation / Bright Dip	____	________	________	______
4	Station 03 — Rinse (post-activation)	____	________	________	______
5	Station 04 — Trivalent Passivate (film, pH/temp, keep it trivalent)	____	________	________	______
6	Bath control — pH / chemistry / metal balance checks	____	________	________	______
7	Station 05 — Rinse (controlled, prep for topcoat)	____	________	________	______
8	**Station 06 — Topcoat / Seal (mandatory step; cobalt awareness)**	____	________	________	______
9	Station 07 — **Dry-Off & Cure (cure per TDS, don't over-bake)**	____	________	________	______
10	Station 08 — Inspect & Hand-Off (color, coverage, salt-spray verdict)	____	________	________	______
11	Passivate + topcoat **system** understanding (no self-healing)	____	________	________	______
12	Keeping it Cr(VI)-free (no oxidizer drag-in / no hex cross-contamination)	____	________	________	______
13	Waste treatment (precipitate metals; no Cr(VI)-reduction step) / drag-out	____	________	________	______
14	LOTO (heaters/pumps/hoists/dryer/ventilation)	____	________	________	______
15	Emergency response — eyewash, shower, spill	____	________	________	______
16	Quality — salt spray / color standard / RoHS hex-free verification	____	________	________	______

Operator name: ________________________ **Hire/start date:** ______________

Supervisor name: ________________________ **Review cycle:** ☐ Annual ☐ Other: __________

⚠ **HEADS-UP**

Safety rows (1, 12, 13, 14, 15) are not optional and not "fill in later." An operator may not work *any* station — and especially not the passivate/topcoat tanks or waste — until the safety/PPE/SDS, keep-it-trivalent, waste/drag-out, LOTO, and emergency rows are signed. Safety competency comes before production competency, always.

PART FOUR — BACK MATTER & CERTIFICATION
20 QUESTIONS · PASSING SCORE 80% (16/20)

TRIVALENT CHROMATE COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

★ **REMEMBER**

Passing score: 80% (16 of 20 correct). Any **safety** question answered wrong (Q5, Q6, Q14, Q15) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: ______________________ **Date:** ____________ **Score:** ________ / 20 **Safety-gate Qs (all correct): 5, 6, 14, 15**

Q1. A trivalent chromate conversion coating forms by:
A. An electric current depositing chromium from a bath onto the part B. A chemical reaction between the freshly plated zinc surface and the process solution (no electric current) C. Spraying molten chromium onto the zinc D. Baking a chromate powder onto the part in a hot oven

Q2. (Short answer) Explain why a trivalent passivate is **NOT self-healing**, and how trivalent finishes still reach high salt-spray performance.

Q3. A trivalent passivate bath typically runs at what **pH**?
A. ~1.8–2.5 (mildly acidic; thick-film types sometimes higher) B. ~7.0 (neutral) C. ~9–10 (mildly alkaline) D. ~13–14 (strongly alkaline)

Q4. (Short answer) Where does the passivate step fit in the overall process, and on which **substrates** is it used?

Q5. [SAFETY GATE] The trivalent passivate bath is best described as:
A. A confirmed human carcinogen requiring an OSHA Cr(VI) program B. A harmless, food-safe rinse C. Not a carcinogen like Cr(VI), but still an acidic chemical bath (skin/eye irritant) that requires PPE and ventilation D. A flammable solvent

Q6. [SAFETY GATE] A specific added hazard of many trivalent **topcoats/sealers** is that they:
A. Contain hexavalent chromium B. Are explosive at room temperature C. Must be stored frozen D. May contain cobalt — a skin/respiratory sensitizer — so check the SDS

Q7. (Short answer) Why is the **topcoat/seal usually mandatory** on a trivalent line (versus optional on the old hex line)?

Q8. Which colors are **standard** for trivalent finishes?
A. Deep hex-style yellow/gold and olive-drab B. Clear/blue-bright and black (with iridescent in between) C. Bright red and orange only D. No color is ever possible

Q9. (Short answer) Where does a trivalent finish's **color** come from, and why is color a weaker signal of protection than it was on hex?

Q10. The main reason a trivalent line applies a topcoat/seal is to:
A. Make the part heavier B. Add electrical conductivity C. Seal the passive film and deliver the corrosion-resistance (salt-spray) performance D. Convert the chromium to hexavalent

Q11. (Short answer) Why must drag-in of oxidizers and cross-contamination be controlled — what could happen if Cr(III) oxidizes to Cr(VI)?

Q12. A trivalent passivate is **not** self-healing because:
A. It contains no mobile hexavalent chromium to leach out and re-protect a scratch B. It is too thick C. It is applied with electricity D. The zinc underneath dissolves the film

Q13. Which regulations are the main reason industry adopted trivalent passivates in place of hexavalent chromate?
A. FDA food-contact rules B. FAA flight rules C. Building fire codes D. RoHS, ELV, and REACH (they restrict hexavalent chromium)

Q14. (Short answer) [SAFETY GATE] List **four** pieces of PPE you must wear at the passivate/topcoat stations, and name **one engineering control** that helps protect you from mist/exposure.

Q15. [SAFETY GATE] Compared to a hexavalent line, trivalent **waste treatment**:
A. Requires a stronger Cr(VI)-reduction step B. Needs no Cr(VI)-reduction step, but you still precipitate/remove the metals (zinc, chromium, cobalt) before discharge C. Lets you pour everything straight down the drain D. Is identical in every way to hexavalent

Q16. To match the corrosion performance of old hex chromate, a trivalent finish relies on:
A. A thicker zinc layer only B. Letting parts rust protectively C. A thicker/uniform passivate PLUS a topcoat/seal and careful process control (the "passivate + topcoat system") D. A hexavalent sealer

Q17. (Short answer) Give **two** honest tradeoffs between trivalent and hexavalent chromate (one advantage of each is fine).

Q18. The everyday spec covering electrodeposited **zinc on steel plus finish Types/Classes (now including trivalent options)** is:
A. ASTM B117 B. MIL-DTL-5541 C. ASTM D3359 D. ASTM B633

Q19. A trivalent chromate conversion coating is applied to:
A. Freshly electroplated **zinc / zinc-alloy (or cadmium)** surfaces B. Bare carbon steel directly C. Anodized aluminum D. Cured powder coat

> ⚠ **SAFETY SECTION — MUST BE CORRECT TO CERTIFY**
> Questions 5, 6, 14, and 15 cover hazards and compliance that can harm you/others or break RoHS. A miss on any safety item means re-teach and re-test before sign-off.

Q20. (Short answer) Name **two** quality checks used to verify a trivalent finish and, in a few words, what each tells you.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

> ⚠ **HEADS-UP**
> Keep this page out of the trainee's copy. Questions **5, 6, 14, and 15** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

Self-check answer distribution (the 12 multiple-choice items): A = 3, B = 3, C = 3, D = 3. If your key clusters on one letter, you've mis-keyed.

Q1. B — A chemical reaction between the zinc surface and the solution; **no electric current**.

Q2. A trivalent passivate **contains no hexavalent chromium**, so there's **no mobile Cr^{6+} to leach out and re-protect a scratch** — it's a **passive barrier**, not self-healing. Trivalent still hits high salt spray as a **system:** a **thicker/uniform passivate + a topcoat/seal** (with corrosion inhibitors) + **careful process control.** Credit for "no mobile Cr^{6+} → passive barrier → topcoat makes up for it."

Q3. A — ~1.8–2.5 (mildly acidic; thick-film/iridescent types are often higher, ~3–4).

Q4. It's the **finishing/passivating step after zinc plating** (the "passivate" the zinc line hands off to). Used on **electroplated zinc, zinc alloys (zinc-nickel/iron/cobalt), and cadmium** — *not* bare steel or aluminum.

Q5. C — Not a carcinogen like Cr(VI), but still an **acidic chemical bath** (skin/eye irritant) requiring PPE and ventilation.

Q6. D — Many trivalent topcoats/sealers may contain **cobalt**, a skin/respiratory **sensitizer** — check the SDS.

Q7. On trivalent, the passivate is a **passive barrier with no self-healing**, so the **topcoat/seal does the load-bearing corrosion work** (seals pores, adds inhibitors/barrier) needed to **meet the salt-spray spec.** On hex the self-healing film already performed, so the seal was optional polish.

Q8. B — Clear/blue-bright and black are standard (iridescent in between); matching the deep hex yellow/gold isn't the goal.

Q9. Color comes from **film type/thickness + the topcoat** (thin-film interference), **not from a leachable chromate.** So unlike hex (where yellow *was* the Cr^{6+} reservoir), trivalent **color doesn't track protection** — judge protection by **salt spray on the topcoated system.**

Q10. C — Seal the passive film and deliver the corrosion-resistance (salt-spray) performance.

Q11. Oxidizer drag-in (e.g., strong oxidizing acids) or hot/high-pH upsets/cross-contamination can **oxidize Cr(III) to Cr(VI)**, which would **re-introduce the carcinogen** *and* **break RoHS compliance** (the part would no longer be hex-free). So keep oxidizers out, hold pH/temp per TDS, and keep hex and trivalent chemistry/equipment separate.

Q12. A — It contains no mobile hexavalent chromium to leach out and re-protect a scratch.

Q13. D — RoHS, ELV, and REACH restrict hexavalent chromium.

Q14. PPE (any four): sealed chemical splash goggles, face shield, chemical-resistant gloves (per SDS, esp. for cobalt), chemical apron/suit, chemical-resistant/non-slip boots, respirator per the program/SDS. **Engineering control (any one):** local exhaust ventilation (and good housekeeping/mist control). (Engineering controls are primary; PPE is the last layer.)

Q15. B — Needs **no Cr(VI)-reduction step**, but you still **precipitate/remove the metals** (zinc, chromium, cobalt) before discharge; metals never go down a drain untreated.

Q16. C — A thicker/uniform passivate **plus** a topcoat/seal and careful process control — the "passivate + topcoat system."

Q17. Any two, e.g.: **Trivalent advantages** — RoHS-compliant, far safer (not a carcinogen), easier waste. **Trivalent disadvantages** — not self-healing, usually needs a topcoat, more steps/cost, fussier baths. **Hex advantages** — cheaper, strong self-healing, best classic corrosion-per-cost. **Hex disadvantages** — carcinogen, RoHS/ELV/REACH-restricted.

Q18. D — ASTM B633 (MIL-DTL-5541 is the *aluminum* chem-film spec; B117 is salt spray; D3359 is paint adhesion).

Q19. A — Freshly electroplated zinc/zinc-alloy (or cadmium) surfaces.

Q20. Any two of: **color/type vs. standard** (right finish — but weak protection signal on trivalent); **coverage/uniformity** (no bare spots); **topcoat integrity / no rub-off** (cured, adherent system); **salt spray ASTM B117** (hours to white/red rust = corrosion life of the system); **paint adhesion ASTM D3359** if painted; **torque-tension** on fasteners; **hex-free verification** for RoHS.

> ★ **REMEMBER**
> 16/20 passes — but every safety question (5, 6, 14, 15) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. We don't gamble with people — even on the safer chromate.

PRINT, FILL IN, SIGN

PLATING POSTERS

PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

CERTIFICATE OF COMPLETION

CHROMATE CONVERSION COATING (TRIVALENT) TRAINING PROGRAM

This certifies that

OPERATOR NAME

has successfully completed the **Trivalent Chromate Conversion Coating** training course — covering all eight process stations (Receive & Inspect, Acid Activation/Bright Dip, Rinse, Trivalent Passivate, Controlled Rinse, Topcoat/Seal, Dry-Off & Cure, and Inspect/Hand-Off), the fundamentals of chemical conversion coating without electric current, why trivalent (Cr III) is the **RoHS/ELV/REACH-compliant** modern replacement for hexavalent chromate, the fact that trivalent is a **passive barrier (not self-healing)** and the **passivate + topcoat system** that delivers its corrosion performance, the clear/blue/iridescent/black color palette and how color differs from hex, drying and curing per the TDS, hex-vs-trivalent positioning, keeping the chemistry **Cr(VI)-free**, the easier (no-Cr(VI)-reduction) metals waste treatment, troubleshooting, and the safety practices for an **acidic, cobalt-bearing but far-lower-hazard** trivalent line — including ventilation, PPE, hygiene, drag-out control, and emergency response — and has passed the Completion Test with a score of ________ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: ____________________ **Re-certification due:** ____________________

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer specifications, and applicable regulatory requirements before production use. Trivalent chromium (Cr III) is the RoHS/ELV/REACH-compliant chromate chemistry and is far lower-hazard than hexavalent chromium, but the process is acidic and some topcoats contain cobalt — always consult current SDS and comply with OSHA, EPA, and local regulations. Keep the chemistry trivalent; do not allow oxidation to, or cross-contamination with, hexavalent chromium.